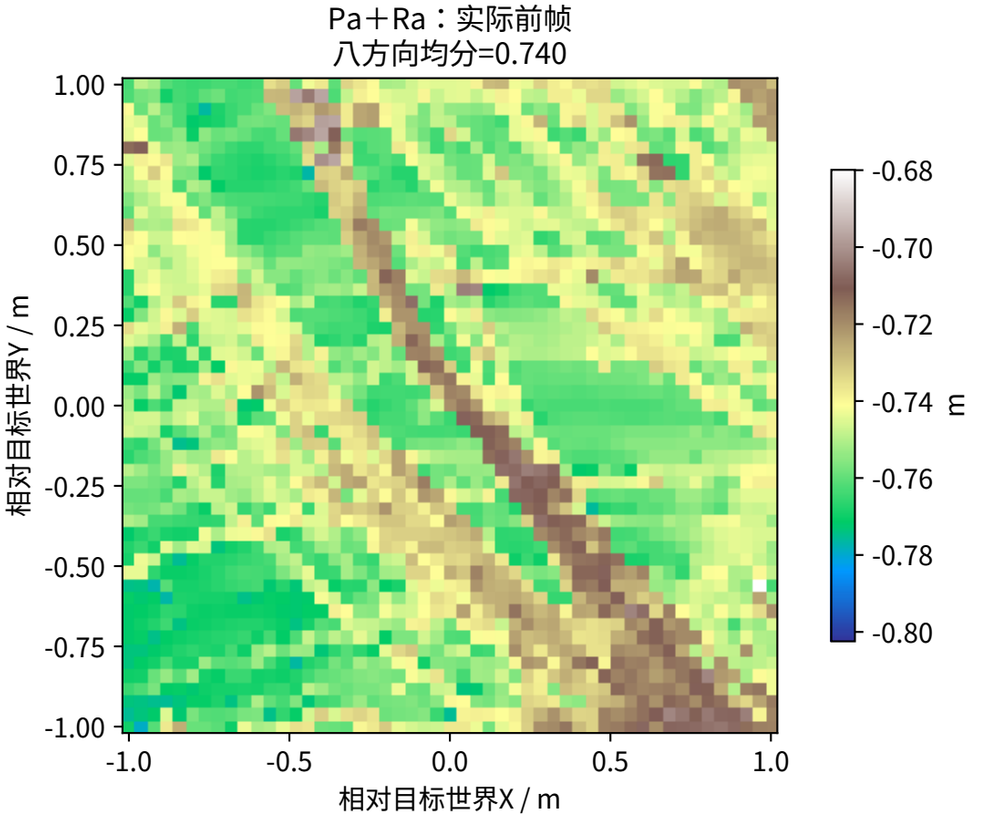
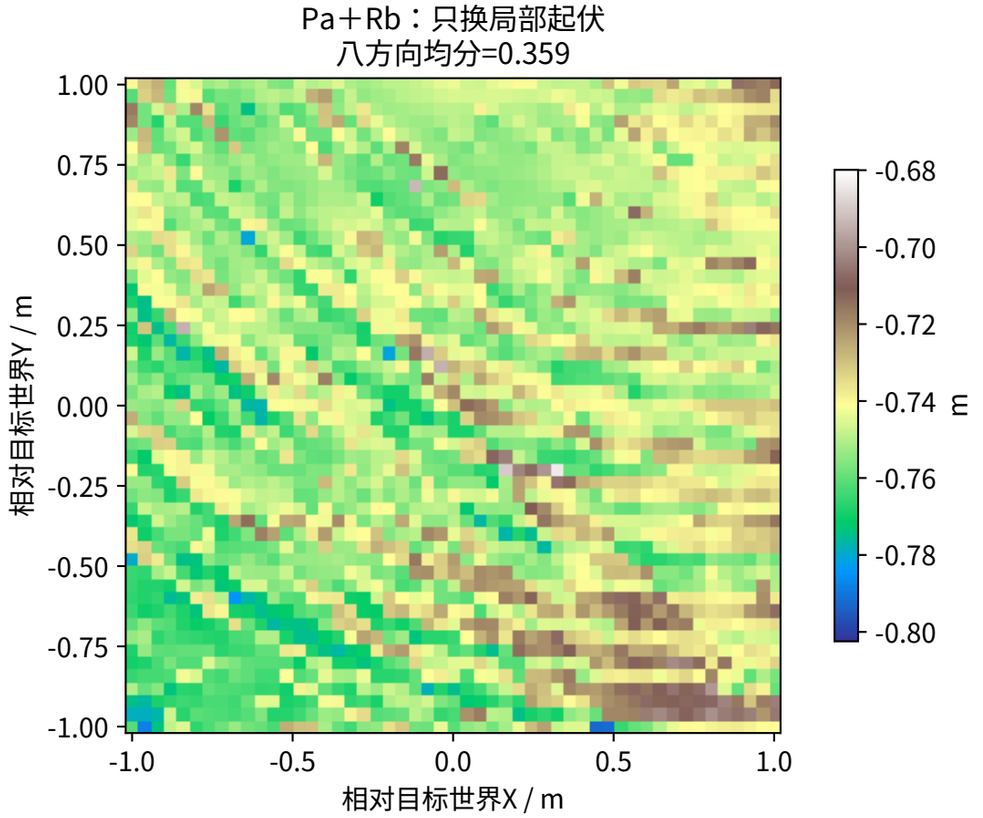
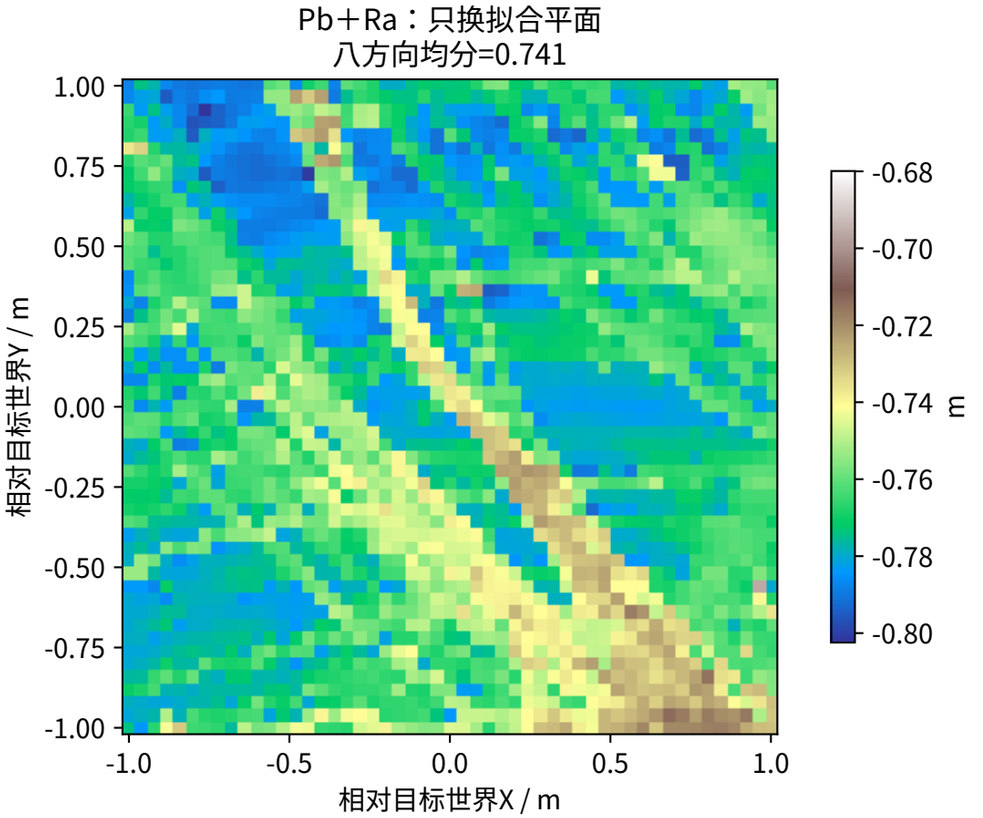
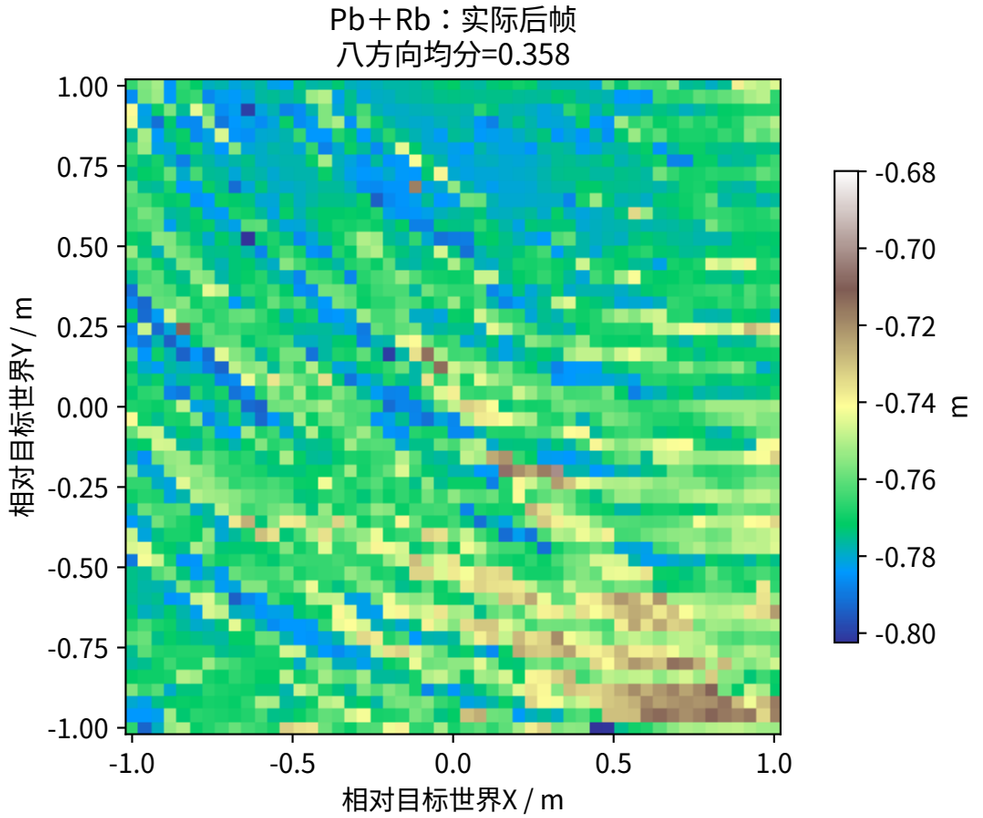
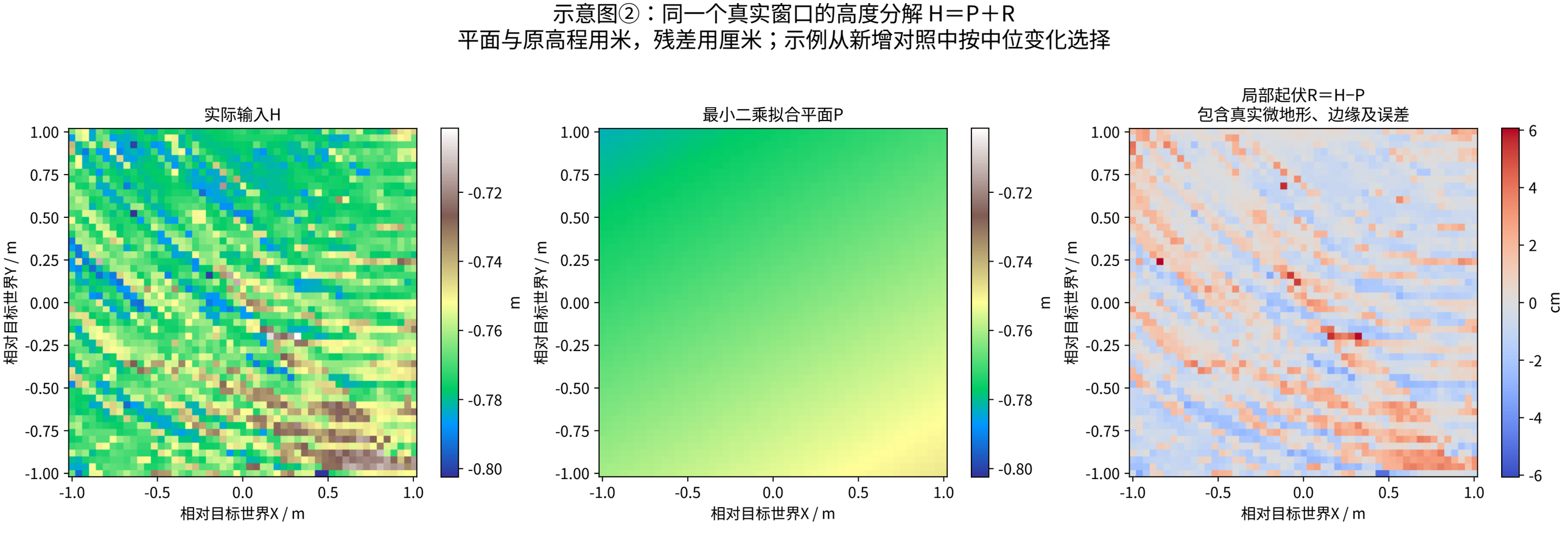
示意图①：完整2×2输入对换，位置/模型/门槛不变
横向改变R，纵向改变P；R是去平面局部起伏，不能预先认定为噪声



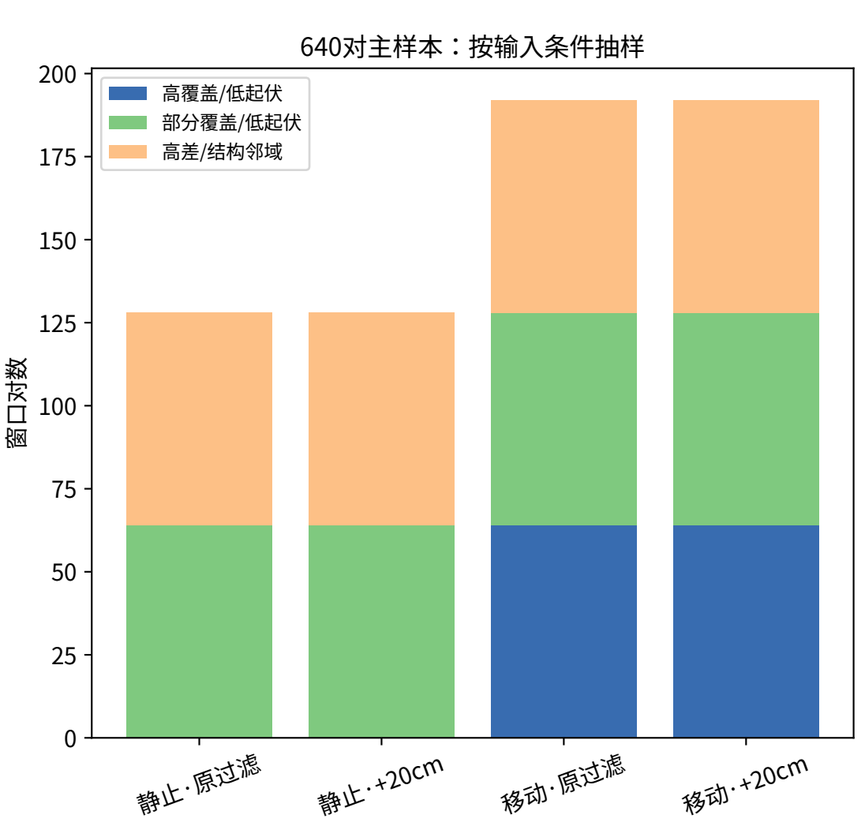


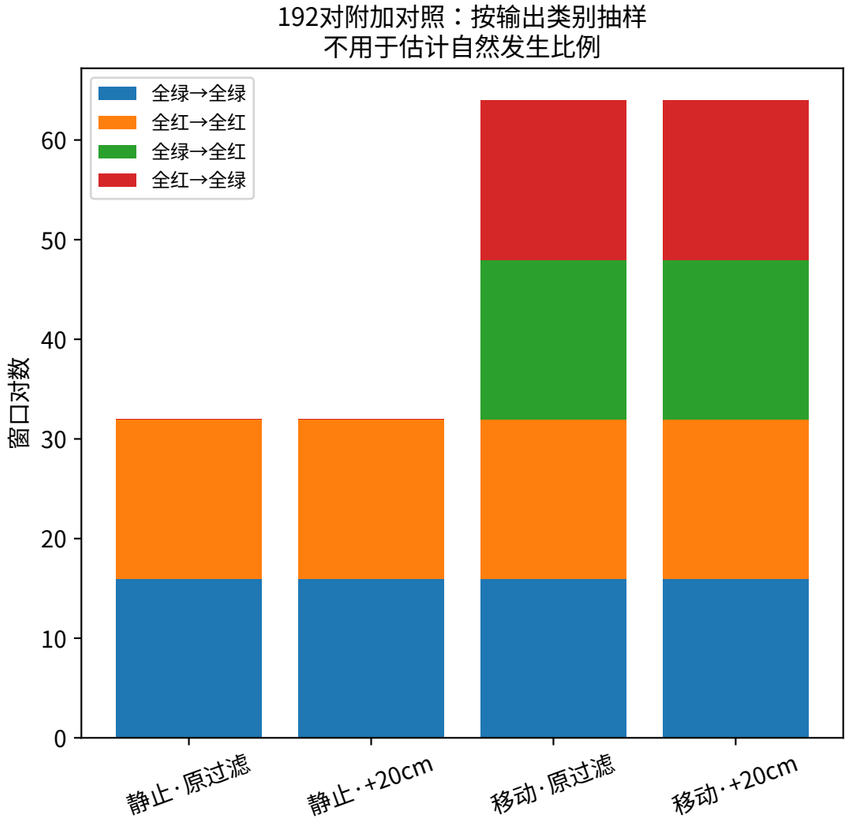


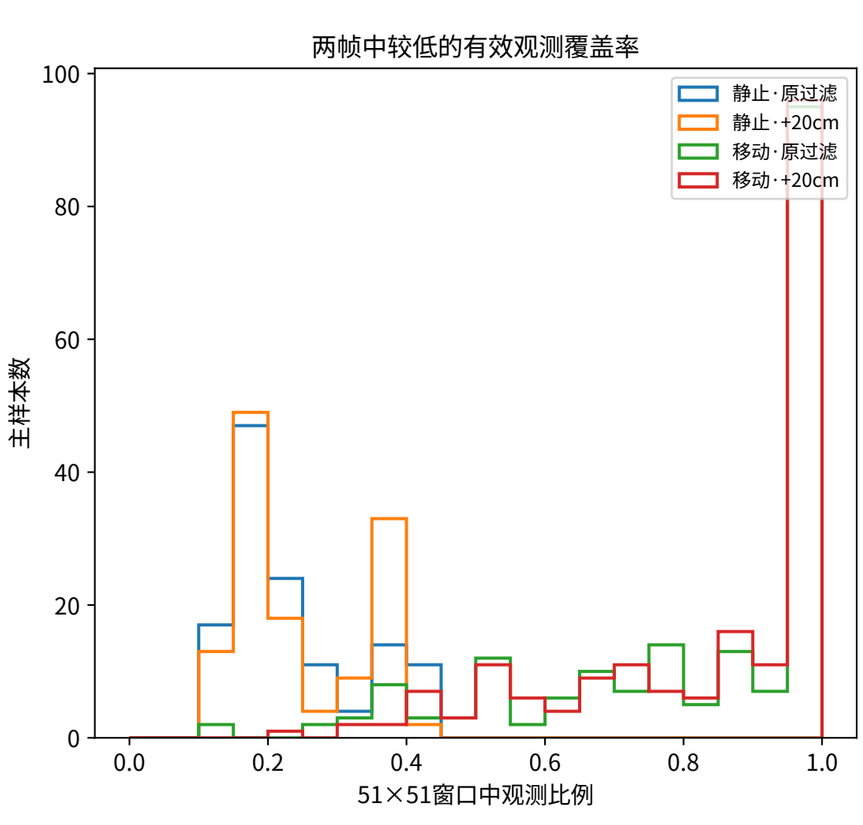



示意图②：同一个真实窗口的高度分解 H＝P＋R
平面与原高程用米，残差用厘米；示例从新增对照中按中位变化选择
实际输入H
最小二乘拟合平面P
局部起伏R＝H−P
包含真实微地形、边缘及误差
相对目标世界X / m
相对目标世界Y / m
m
cm
1.00
0.75
0.50
0.25
0.00
-0.25
-0.50
-0.75
-1.00
-0.5
0.0
0.5
1.0
-0.72
-0.74
-0.76
-0.78
-0.80
6
4
2
0
-2
-4
-6

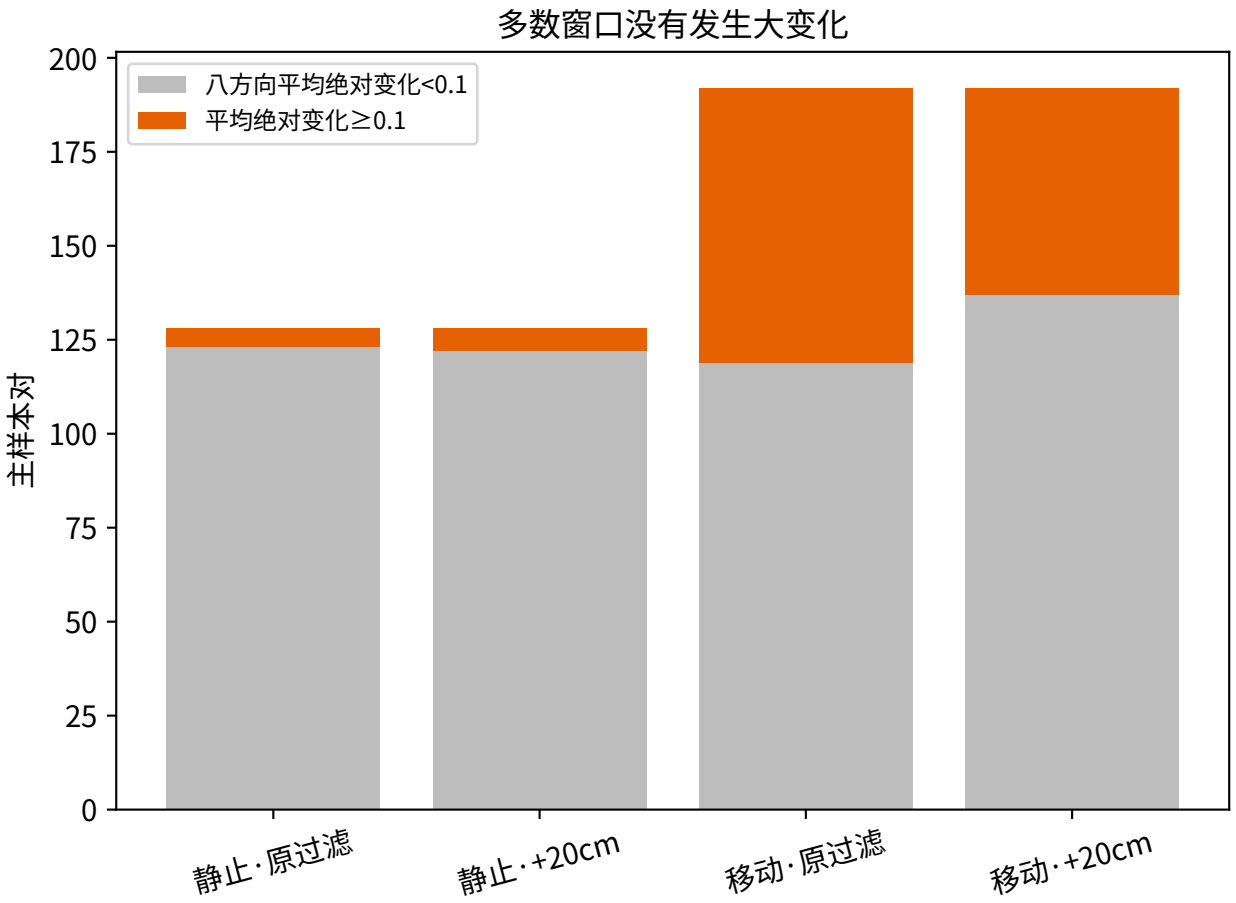
抽样覆盖：2份真实录包 × 2种处理；四个版本不是四次独立实车实验
静止包没有符合≥98%覆盖的低起伏窗口；如实留空，没有人为补齐

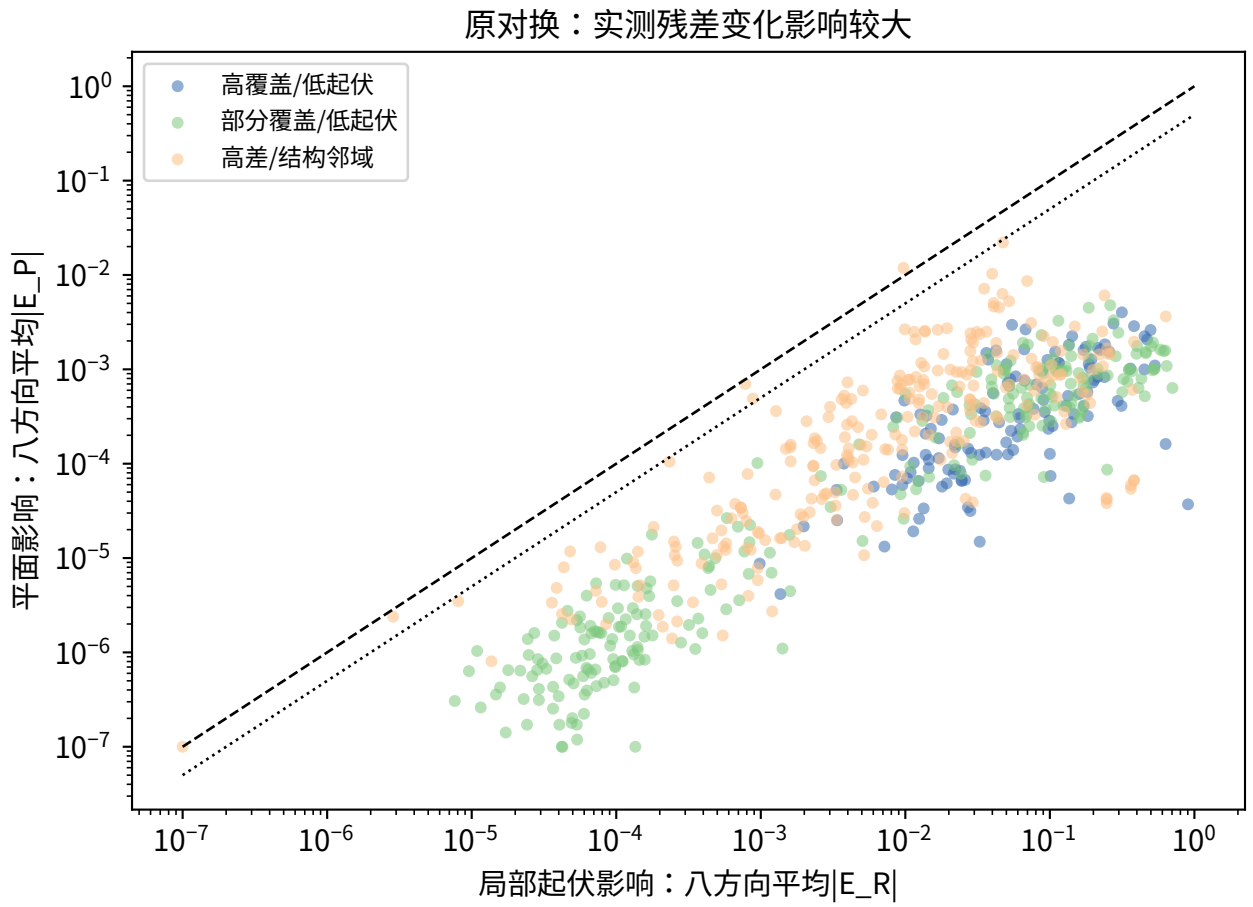


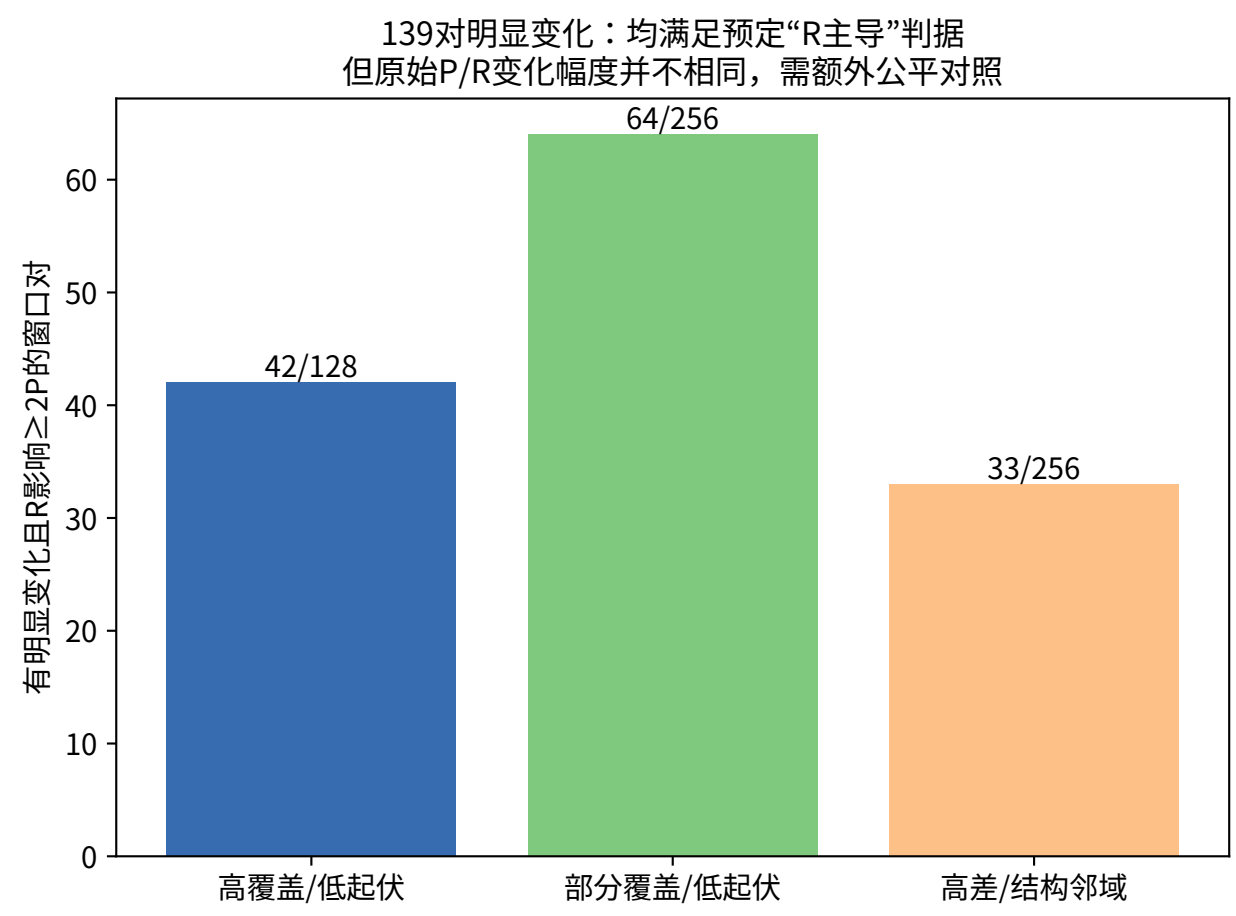


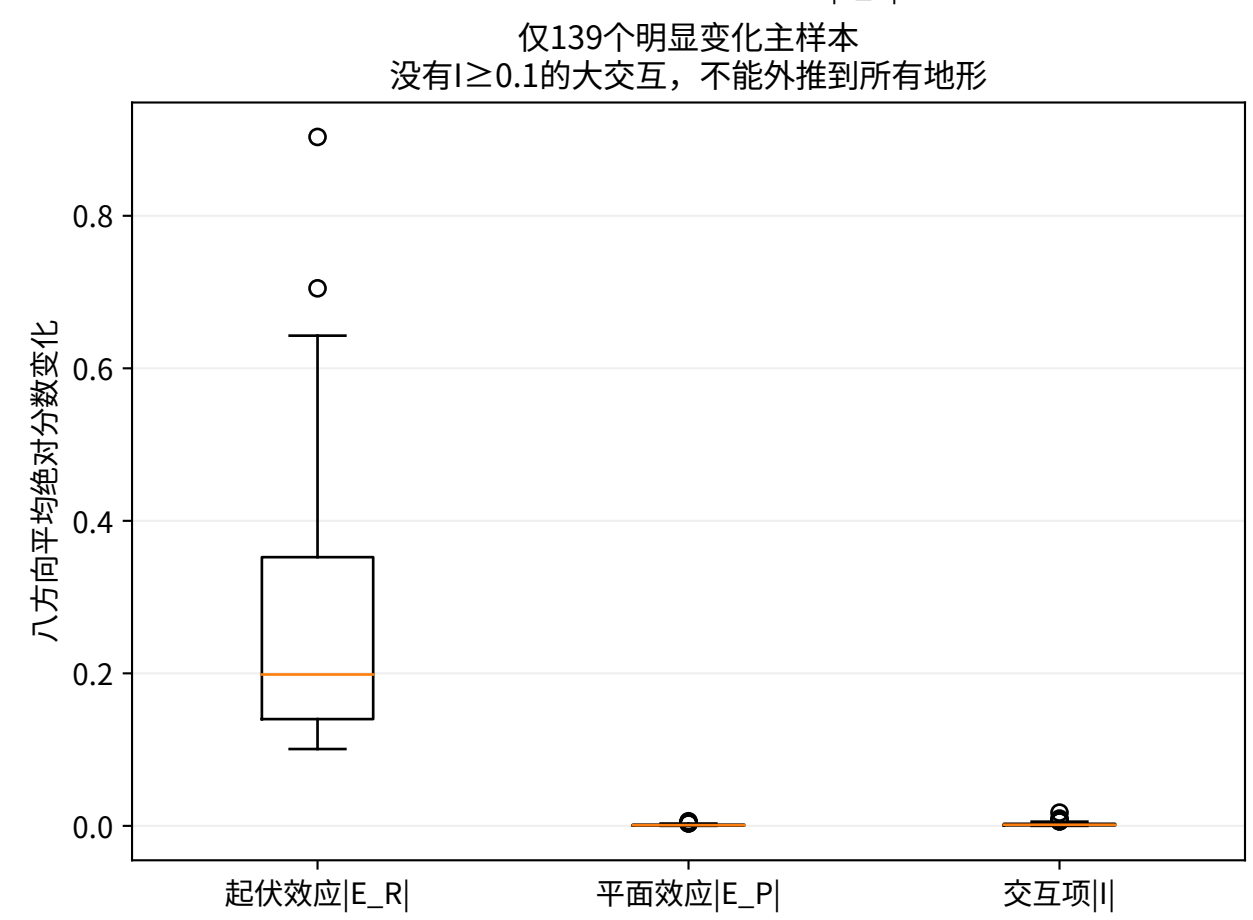


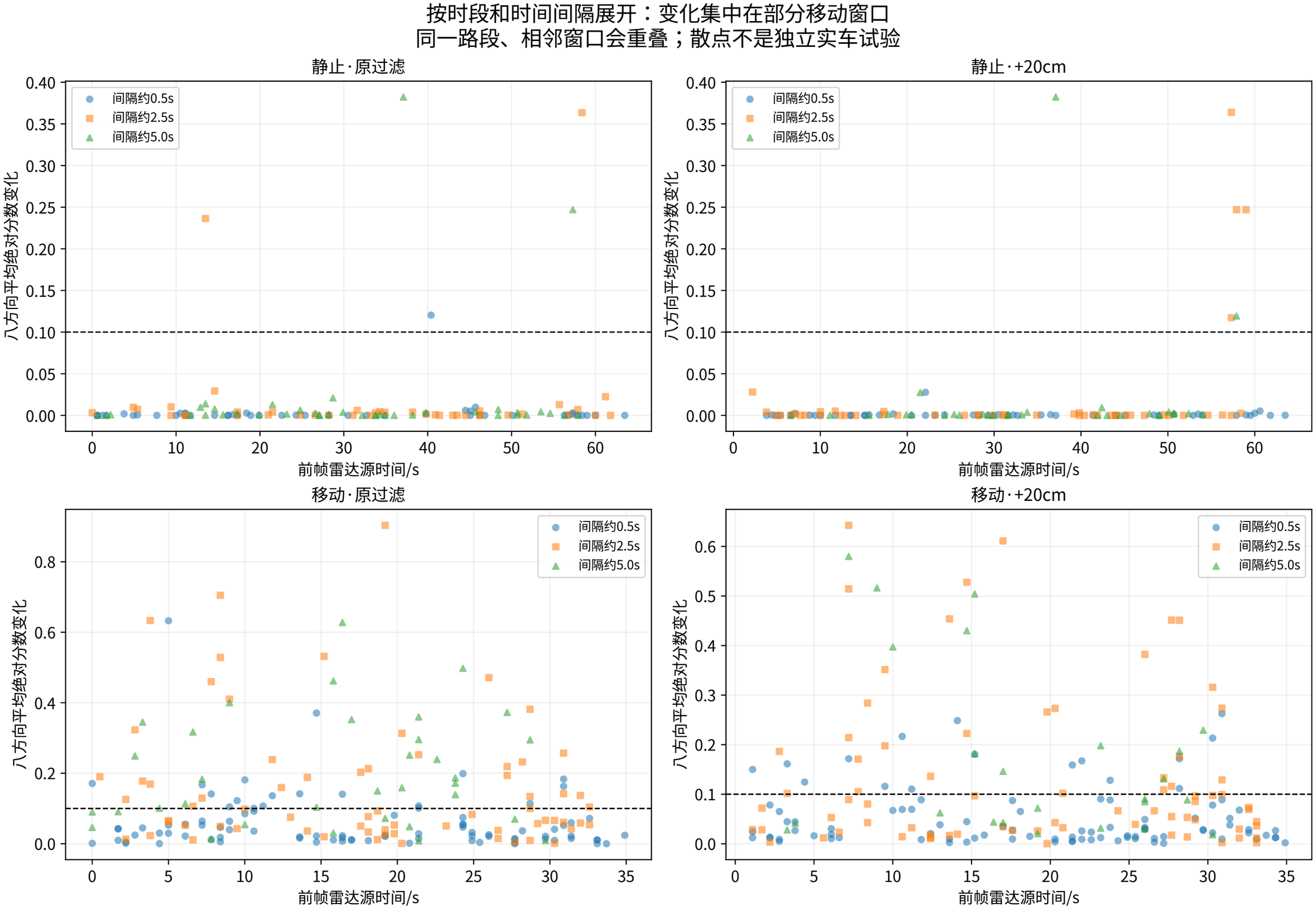
主结果：支持“本次窗口变化主要随局部起伏走”
不等于证明这些起伏是噪声，也不等于证明真实地面不可通过

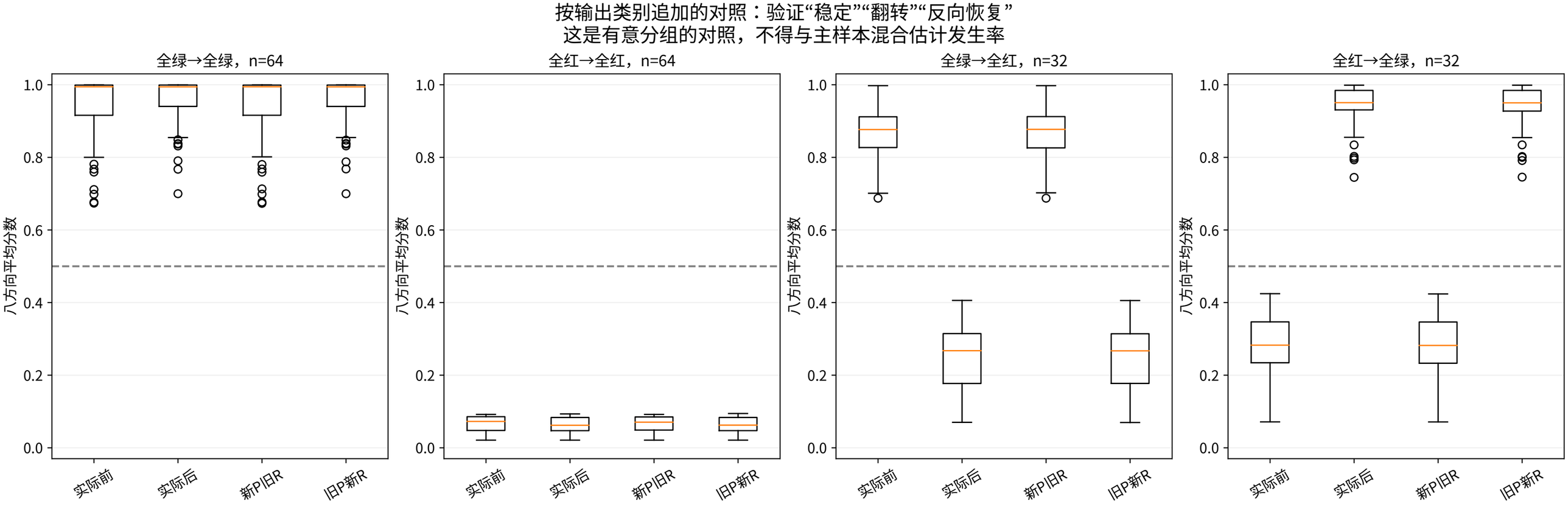


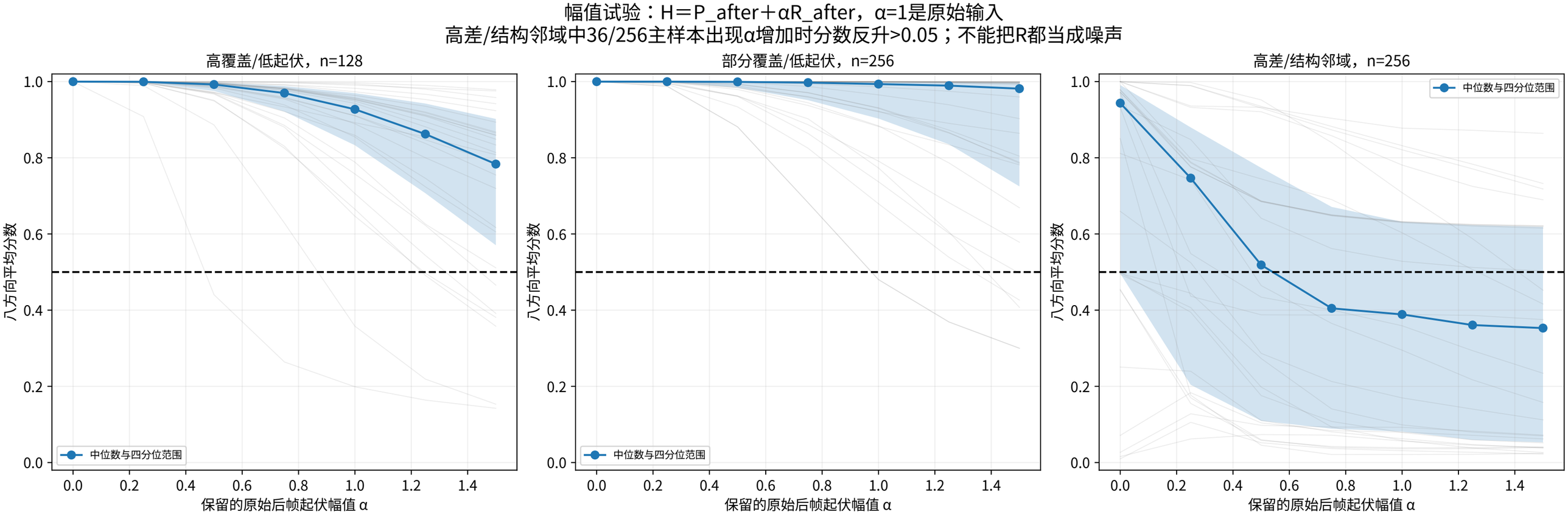


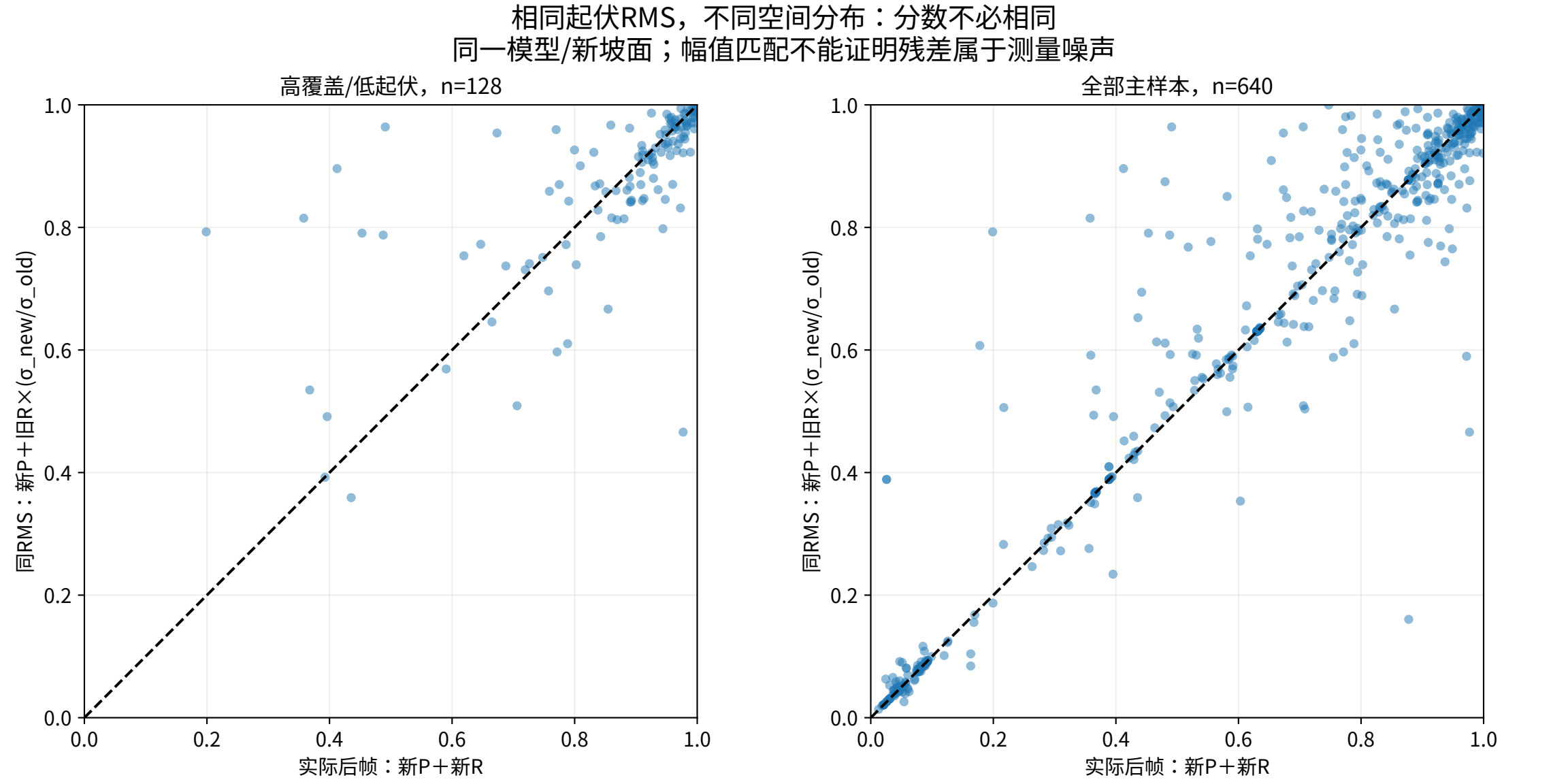


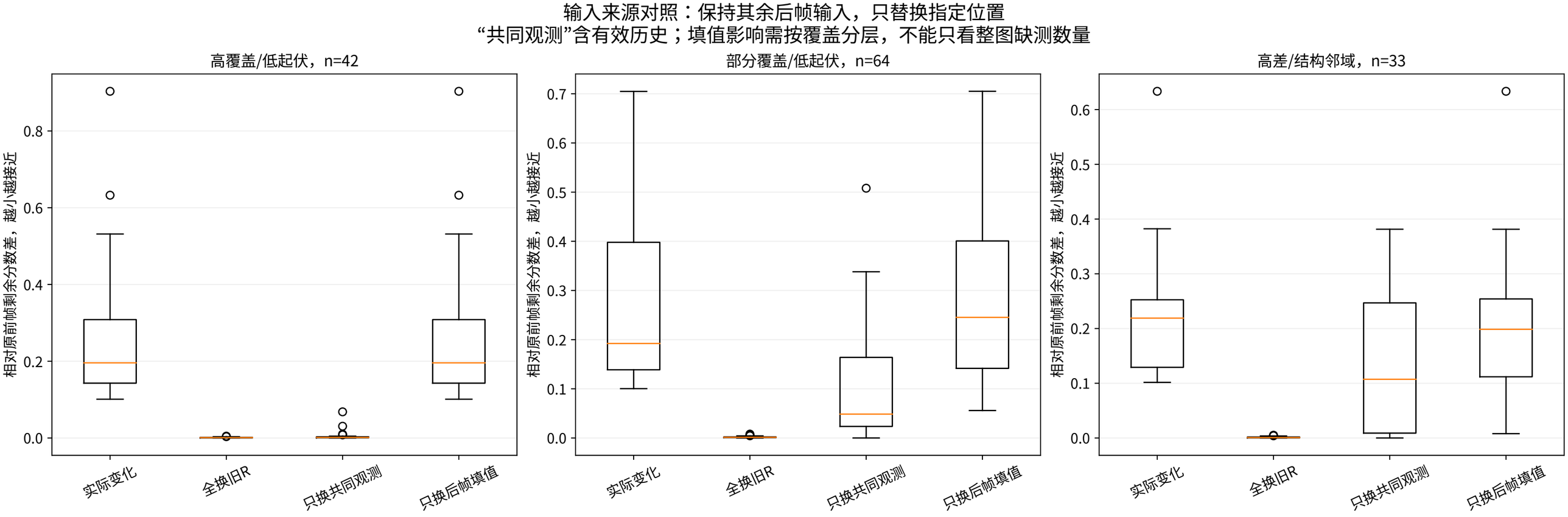



按时段和时间间隔展开：变化集中在部分移动窗口
同一路段、相邻窗口会重叠；散点不是独立实车试验
静止·原过滤
静止·+20cm
移动·原过滤
移动·+20cm
间隔约0.5s
间隔约2.5s
间隔约5.0s
八方向平均绝对分数变化
前帧雷达源时间/s

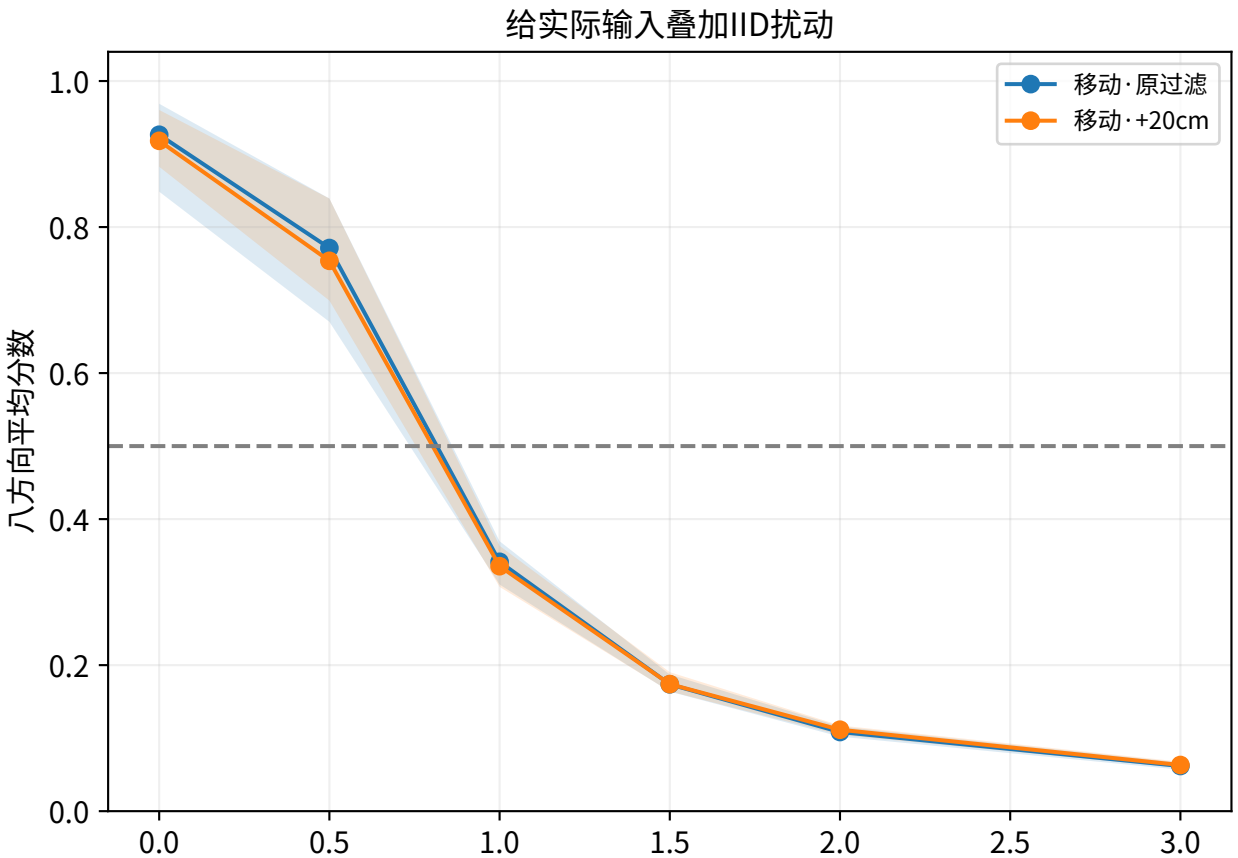

按输出类别追加的对照：验证“稳定”“翻转”“反向恢复”
这是有意分组的对照，不得与主样本混合估计发生率
全绿→全绿，n=64
全红→全红，n=64
全绿→全红，n=32
全红→全绿，n=32
八方向平均分数
1.0
0.8
0.6
0.4
0.2
0.0
实际前
实际后
新P旧R
旧P新R

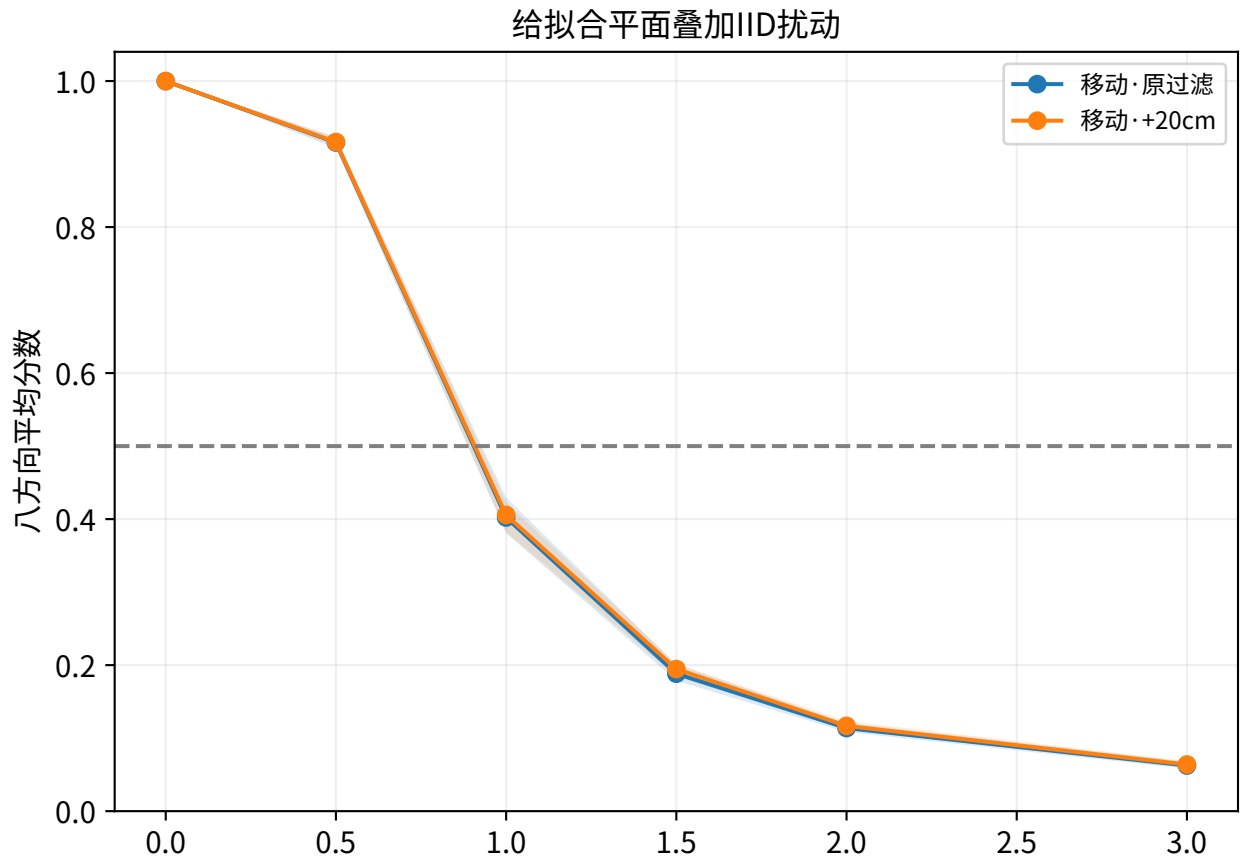

幅值试验：H＝P_after＋αR_after，α=1是原始输入
高差/结构邻域中36/256主样本出现α增加时分数反升>0.05；不能把R都当成噪声
高覆盖/低起伏，n=128
部分覆盖/低起伏，n=256
高差/结构邻域，n=256
八方向平均分数
保留的原始后帧起伏幅值 α
中位数与四分位范围
0.0
0.2
0.4
0.6
0.8
1.0
1.2
1.4

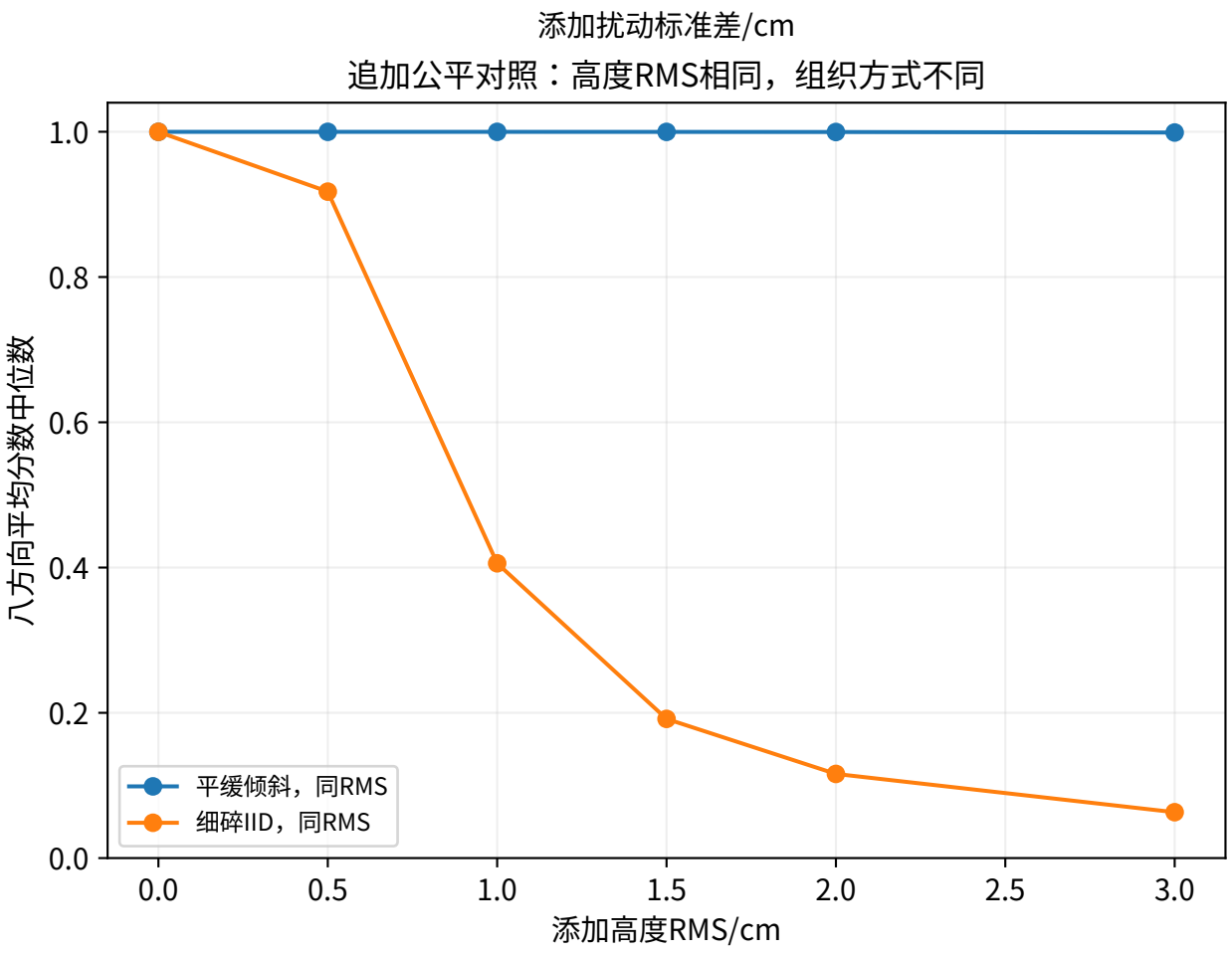

相同起伏RMS，不同空间分布：分数不必相同
同一模型/新坡面；幅值匹配不能证明残差属于测量噪声
高覆盖/低起伏，n=128
全部主样本，n=640
同RMS：新P＋旧R×(σ_new/σ_old)
实际后帧：新P＋新R
0.0
0.2
0.4
0.6
0.8
1.0

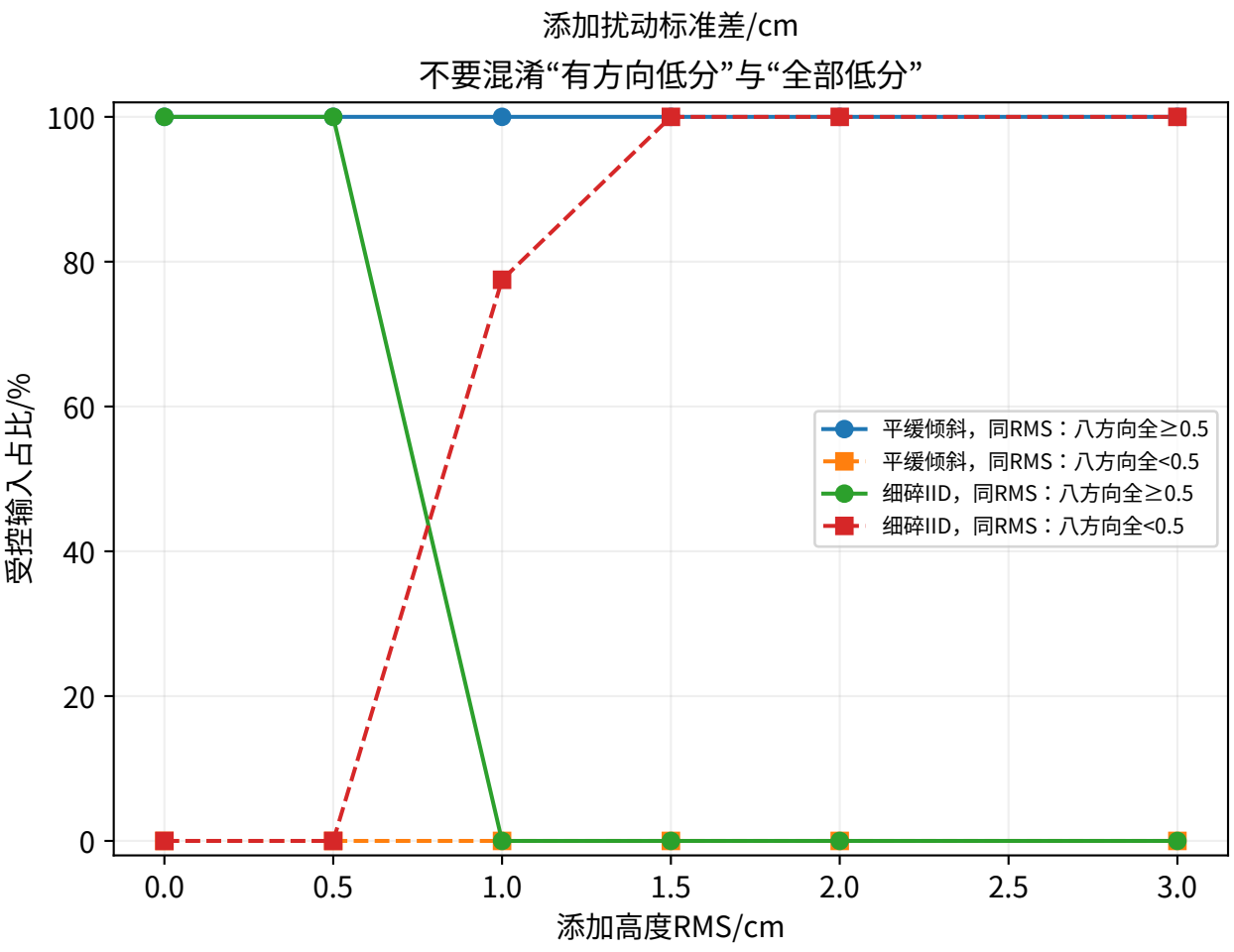

输入来源对照：保持其余后帧输入，只替换指定位置
“共同观测”含有效历史；填值影响需按覆盖分层，不能只看整图缺测数量
高覆盖/低起伏，n=42
部分覆盖/低起伏，n=64
高差/结构邻域，n=33
相对原前帧剩余分数差，越小越接近
实际变化
全换旧R
只换共同观测
只换后帧填值

受控扰动：32个记录窗口＋5种扰动种子，不是1920次实车试验
IID未证明等同真实雷达误差；同RMS对照为看完主结果后预先冻结的追加试验

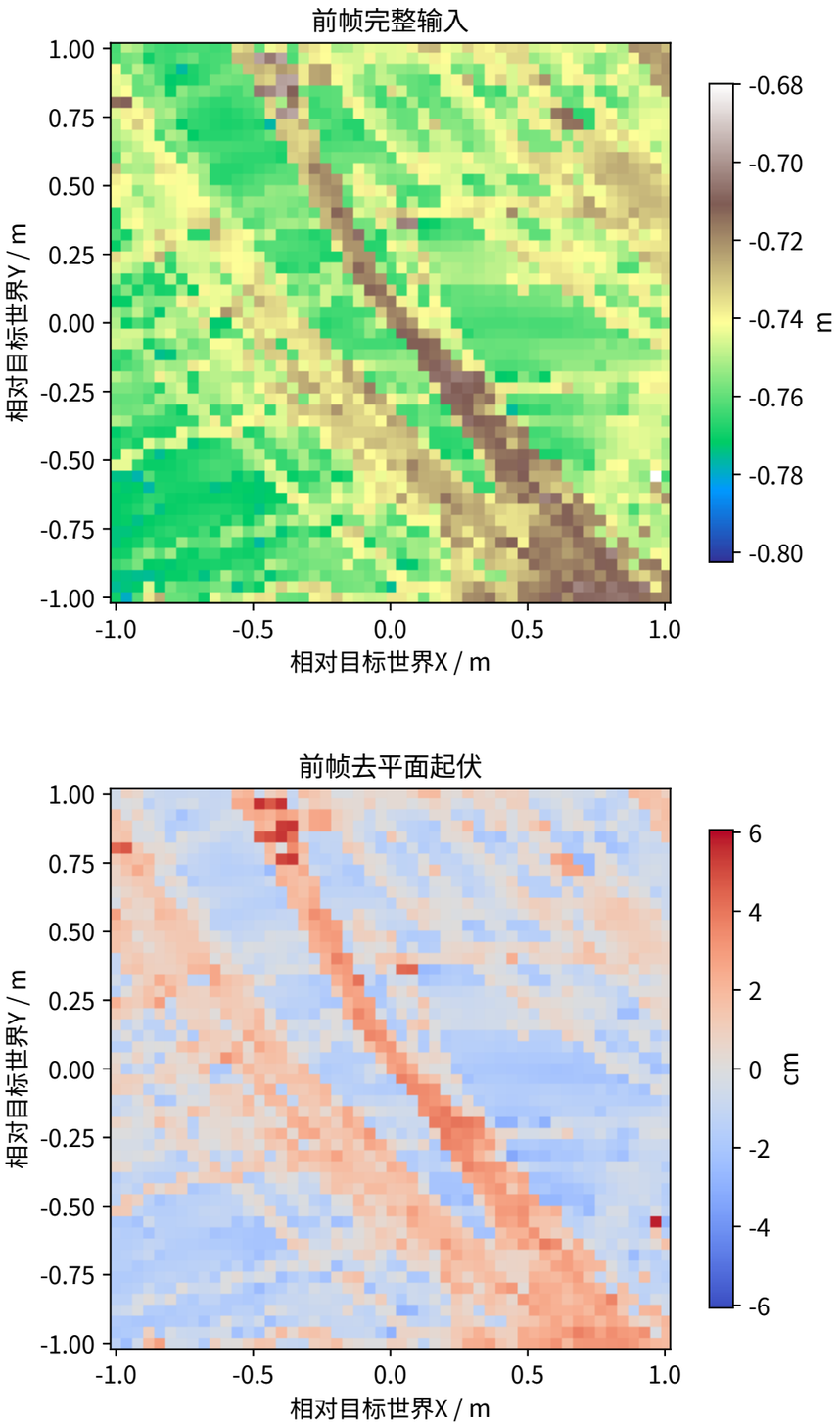


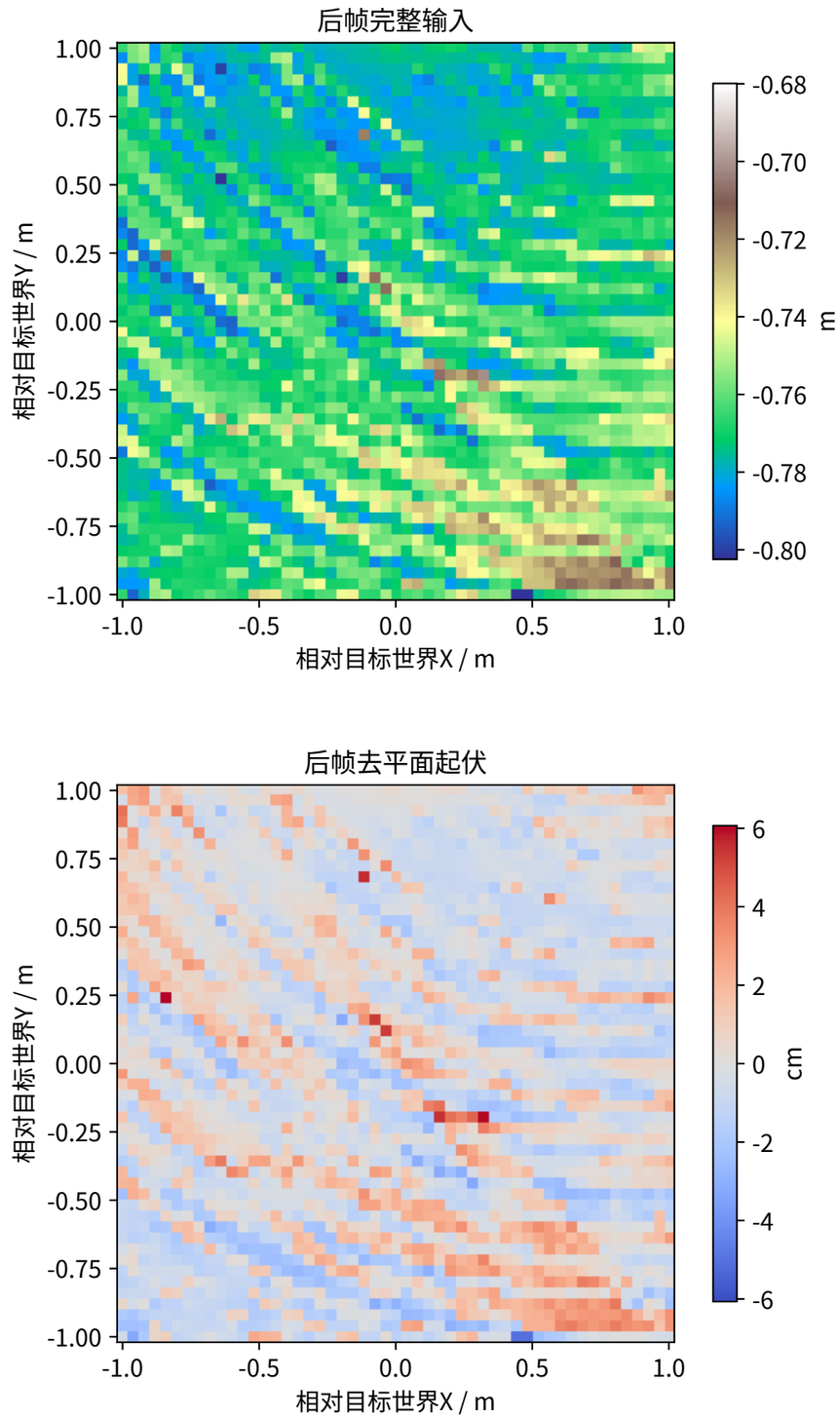


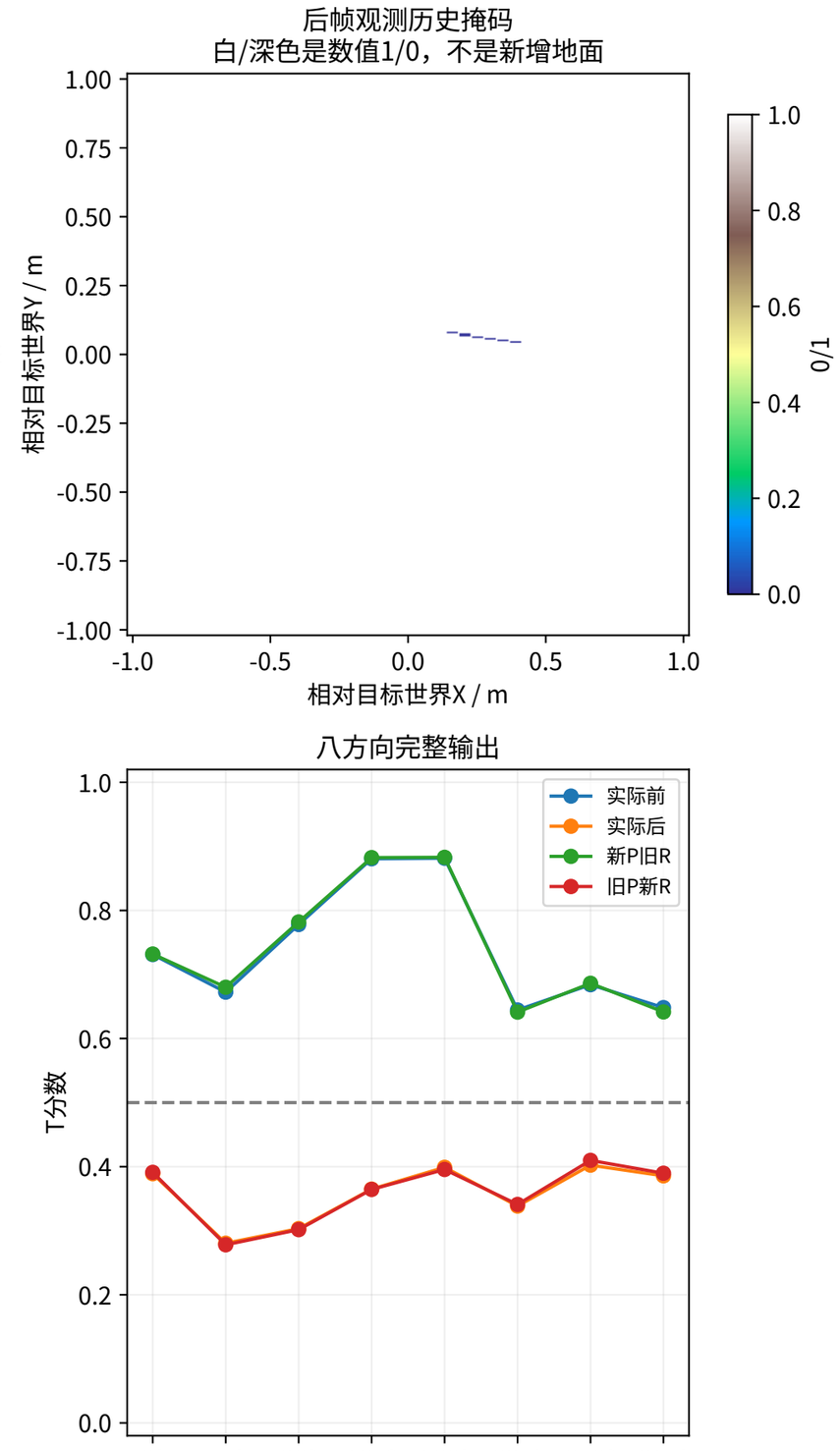




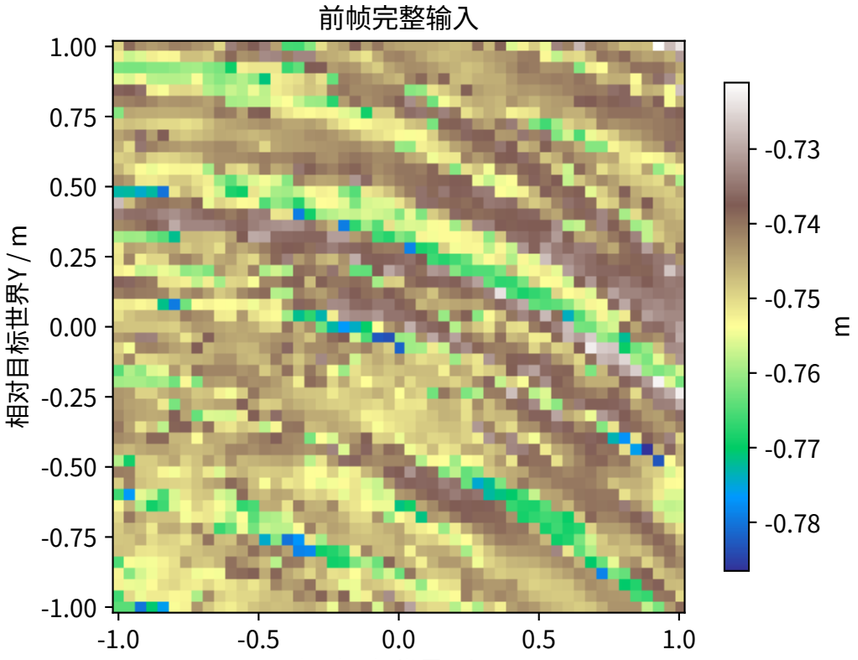
对照：全绿→全红｜样本#627，plus20_moving
世界(-3,13)m，26.00→28.70s，覆盖≥98.7%，按组内中位变化选取

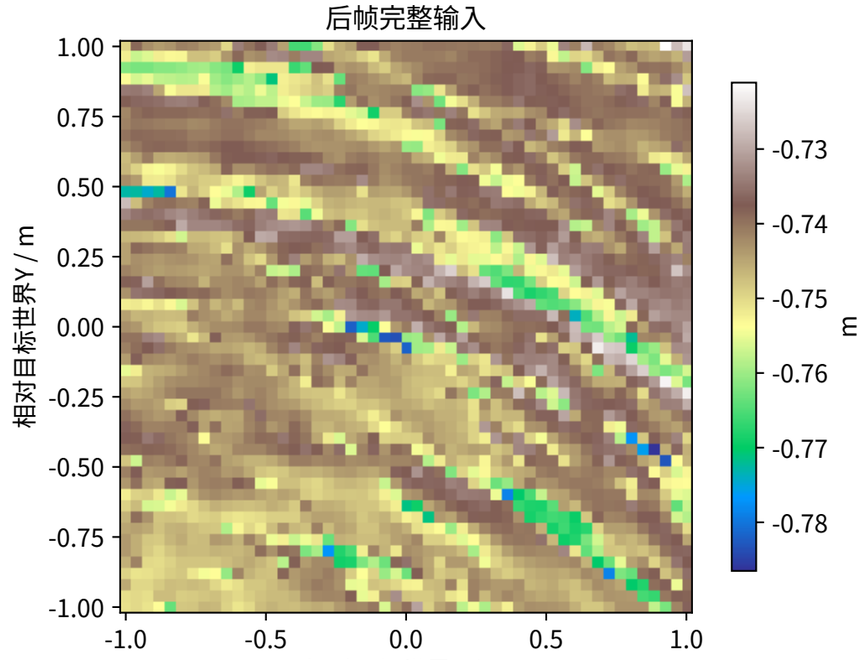


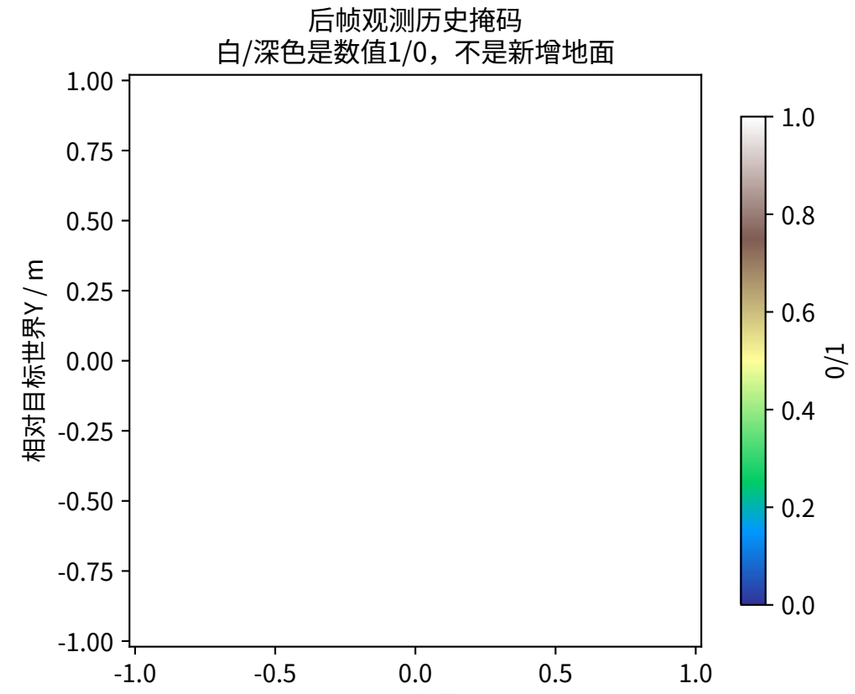


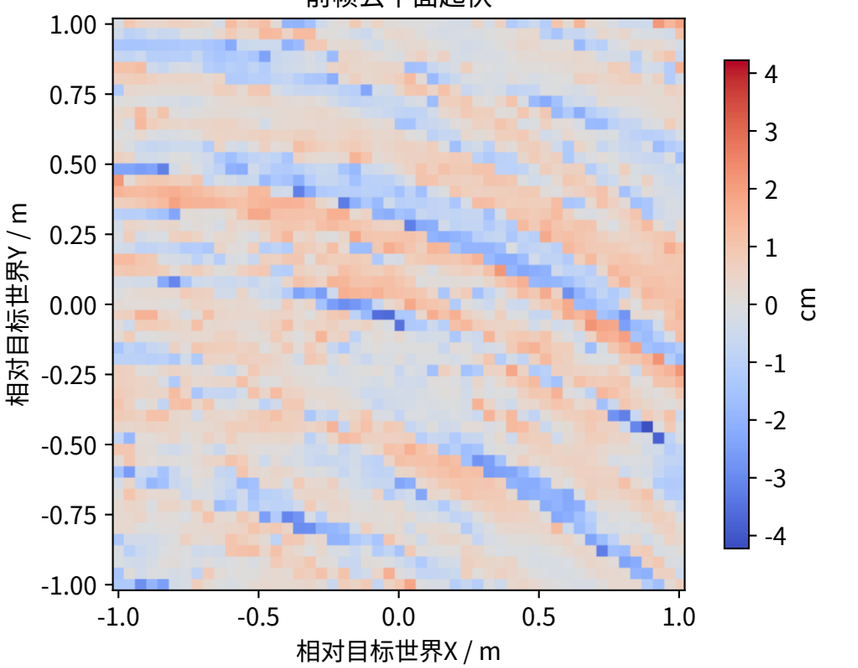


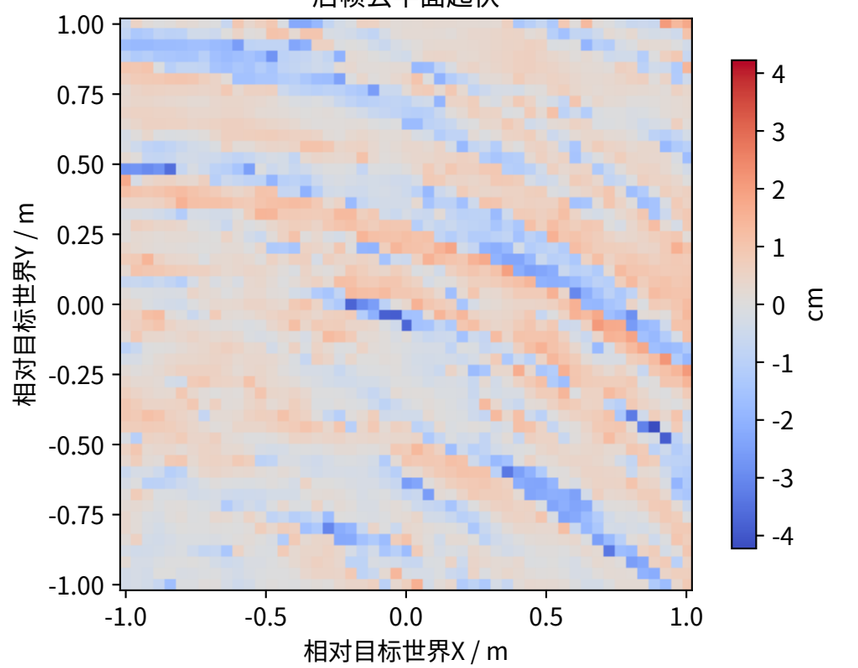
对照：始终全绿｜样本#581，plus20_moving
世界(-1,18)m，32.60→33.10s，覆盖≥100.0%，按组内中位变化选取

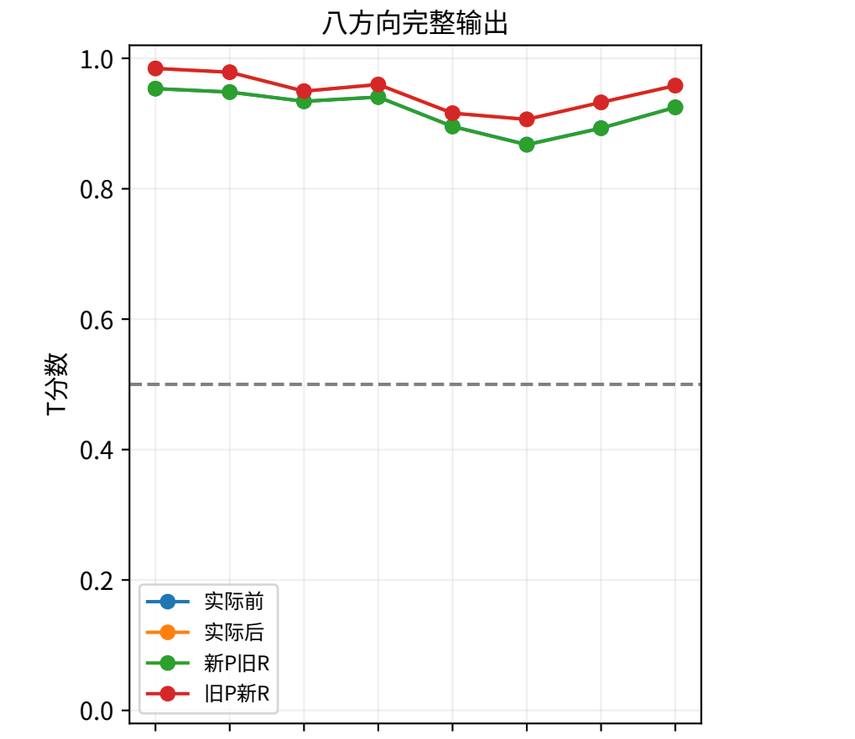




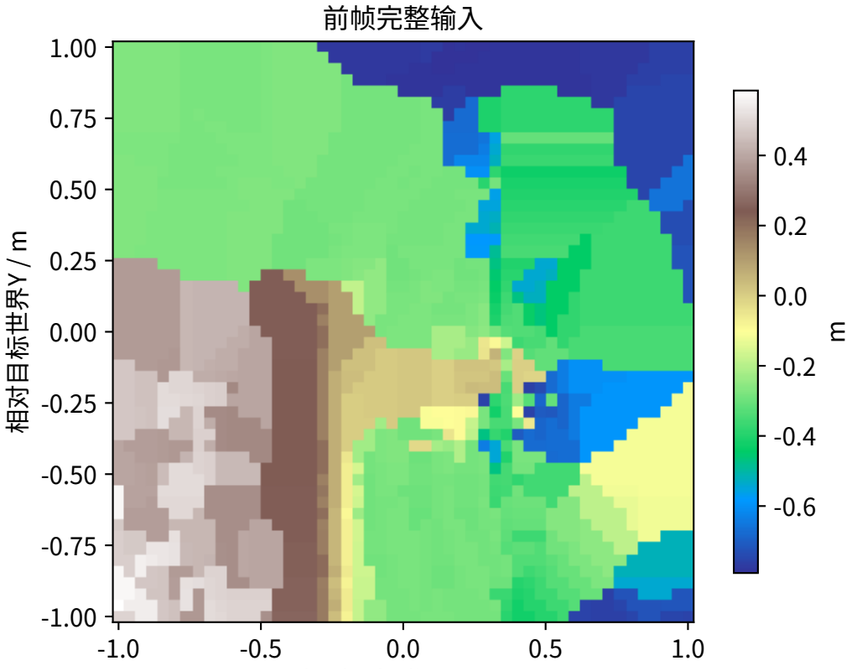


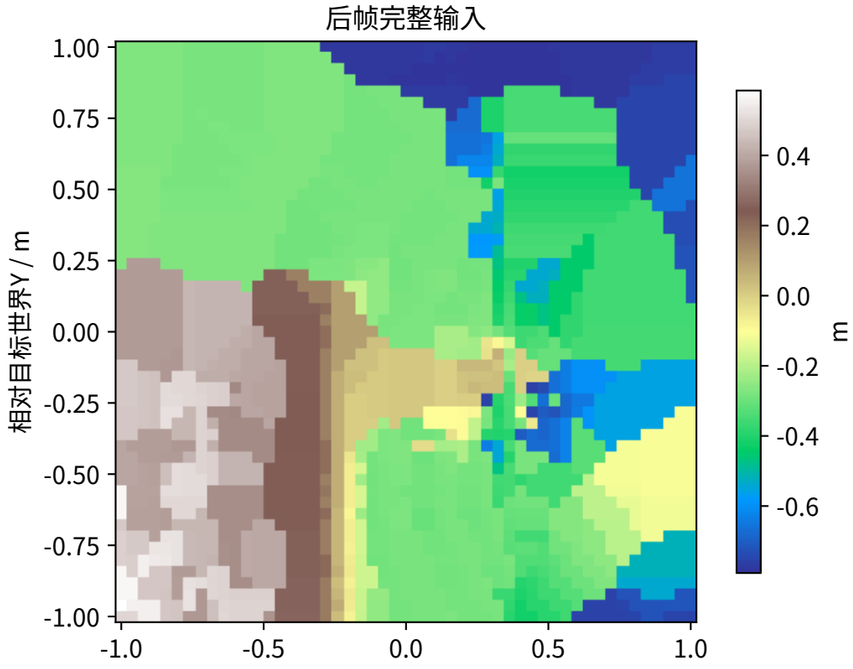


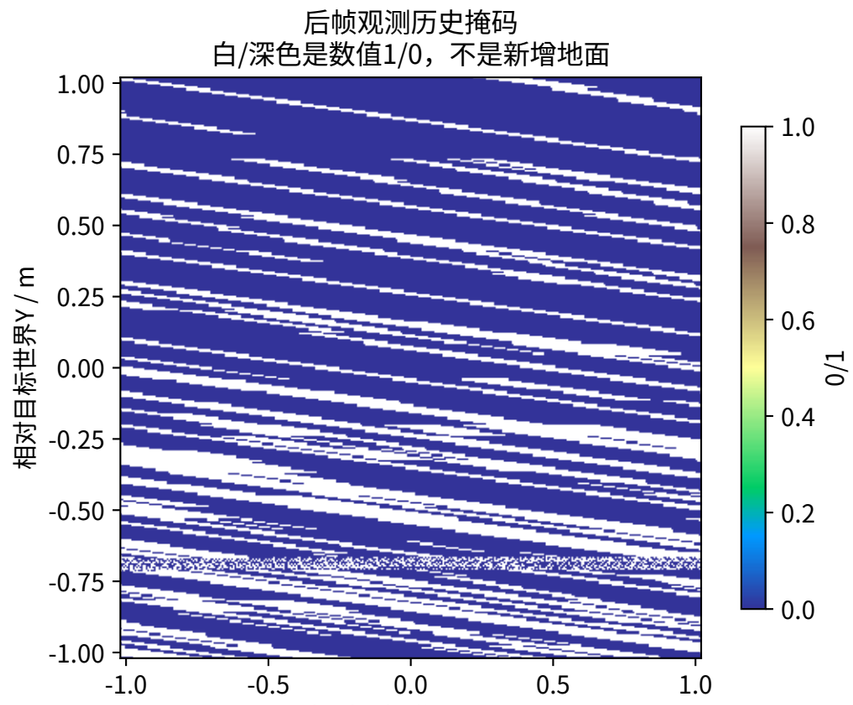


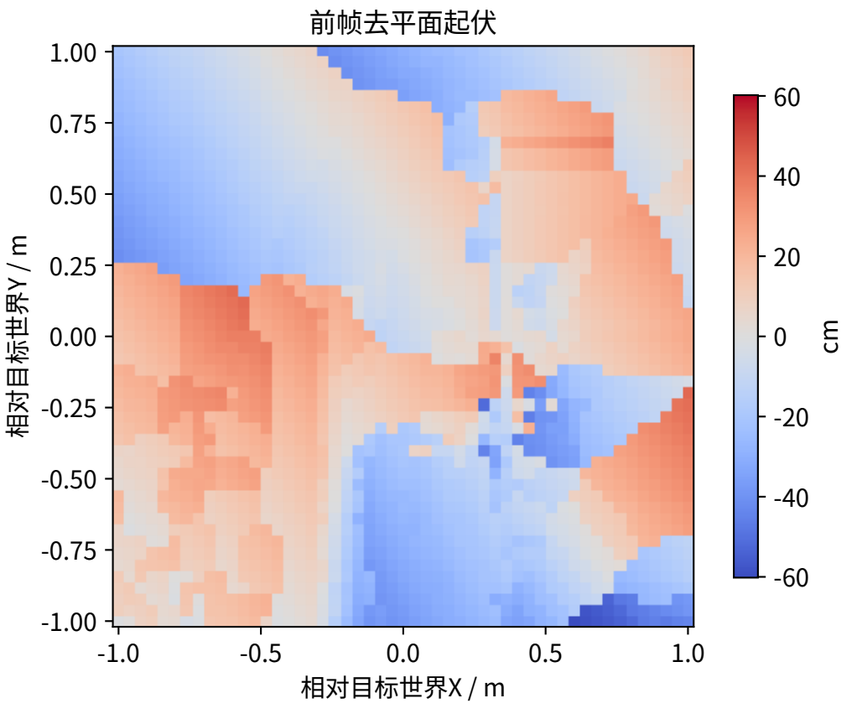


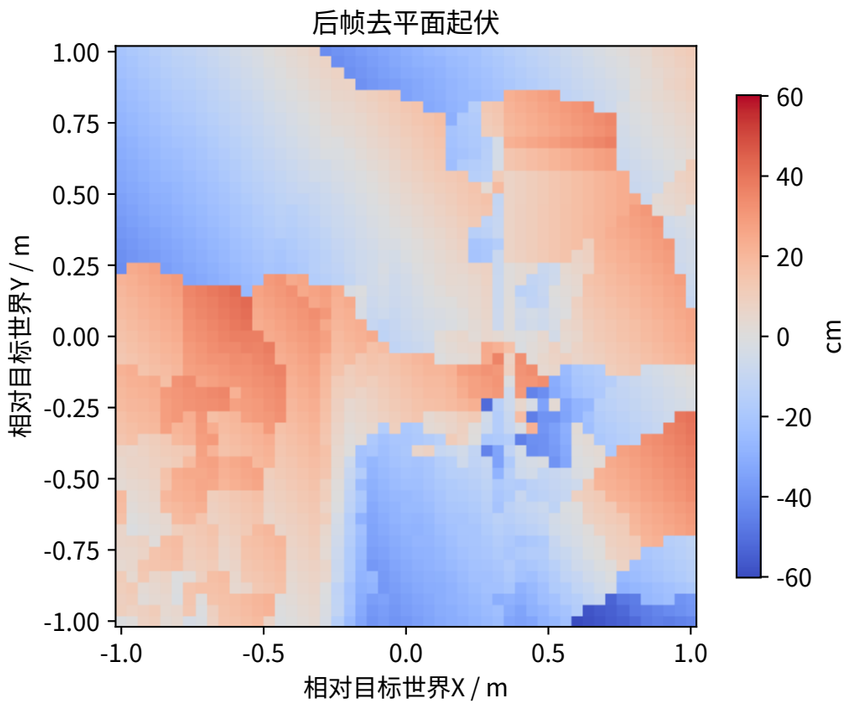
对照：始终全红｜样本#147，original_stationary
世界(-4,17)m，8.80→11.10s，覆盖≥25.1%，按组内中位变化选取

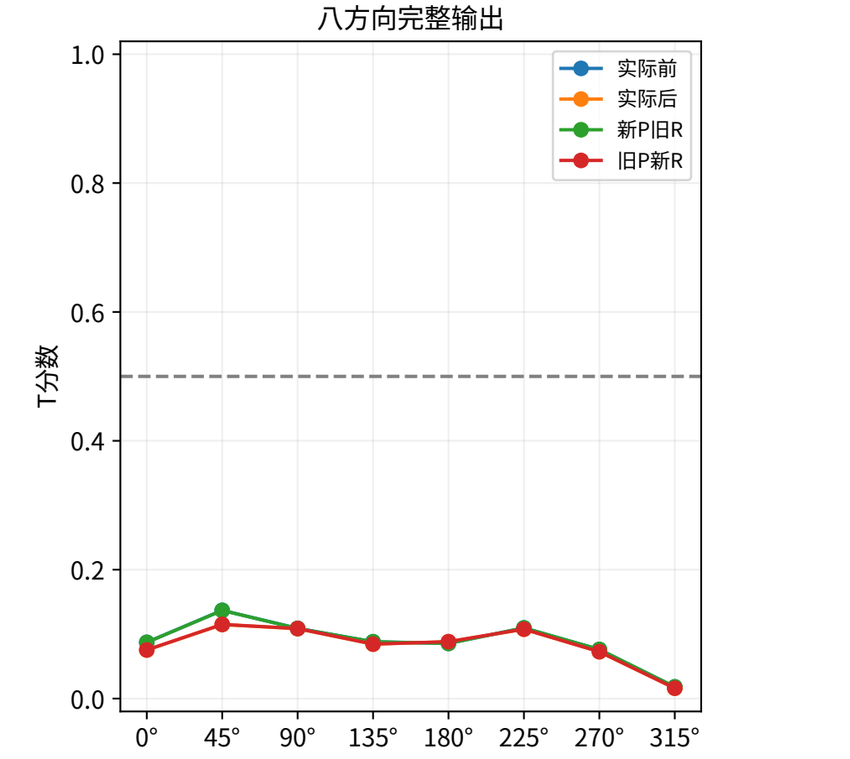




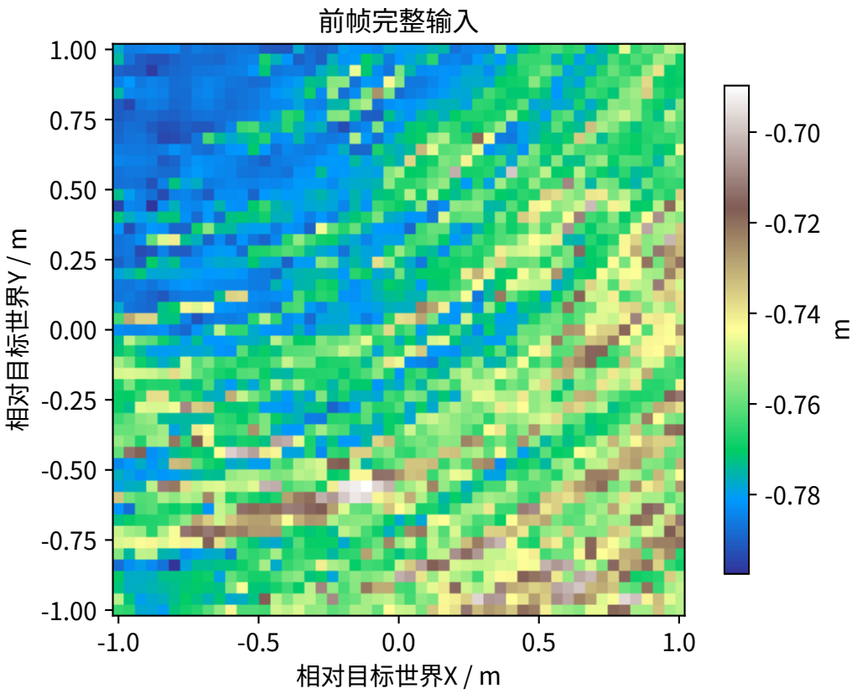


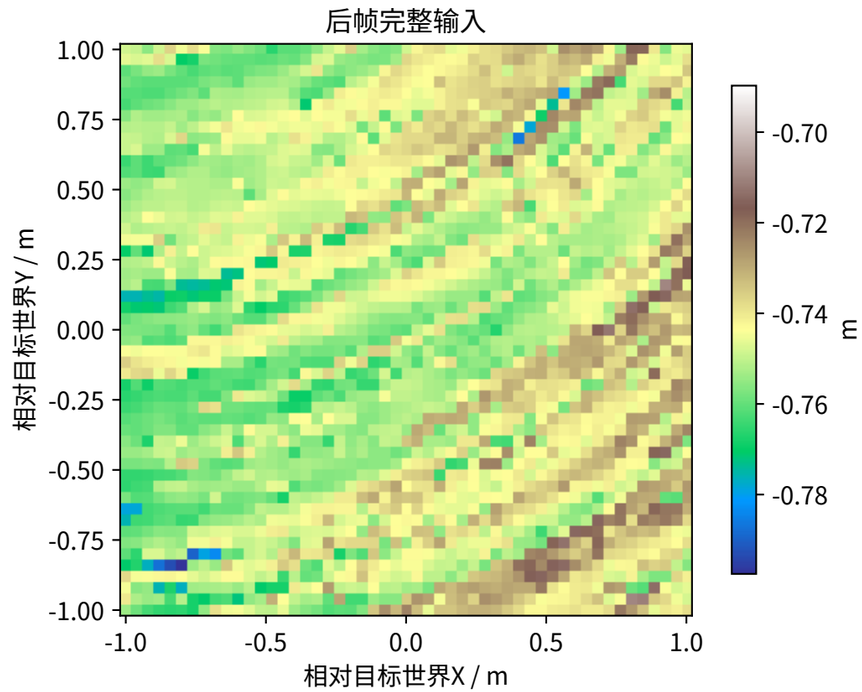


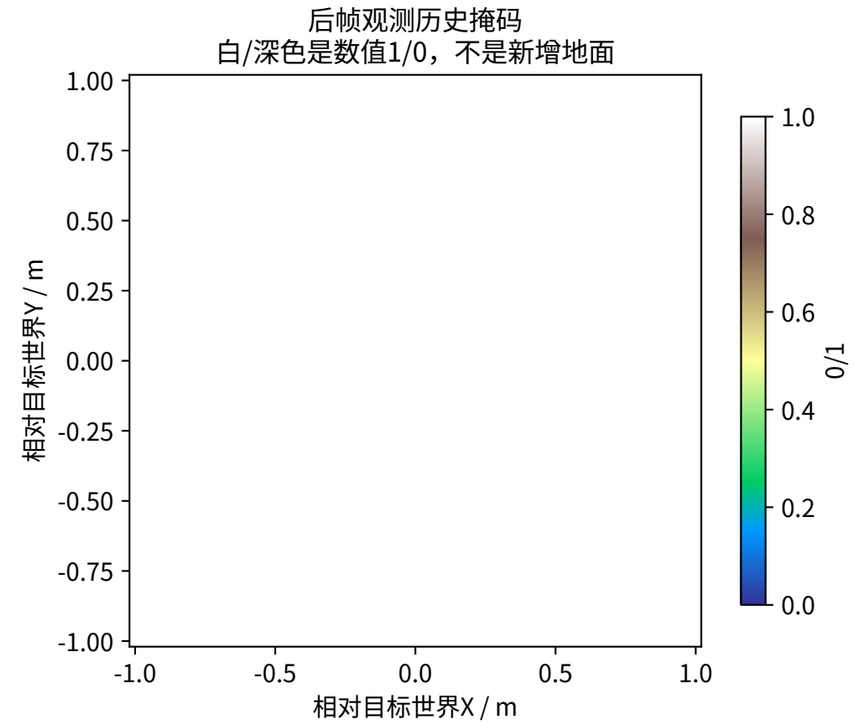


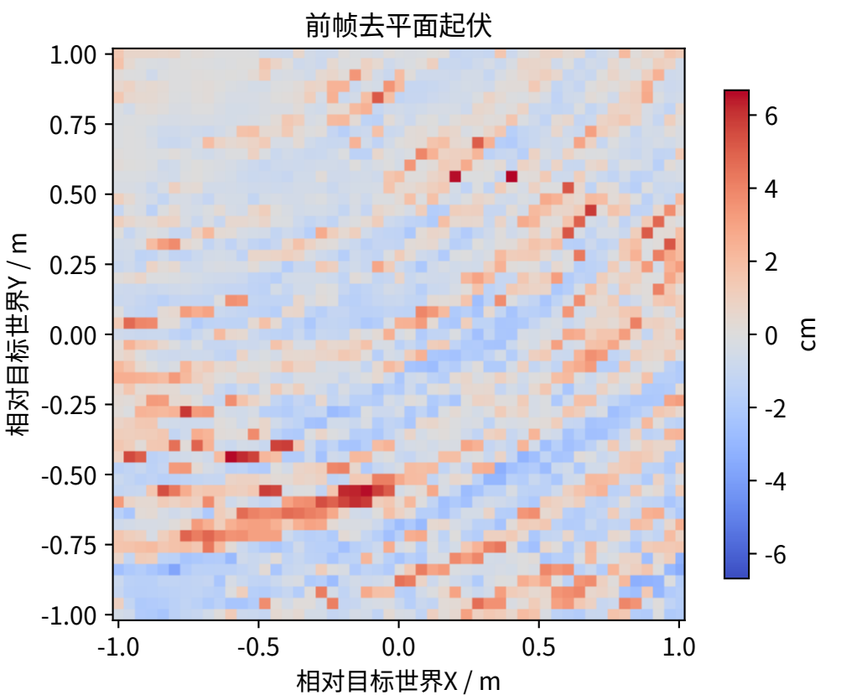


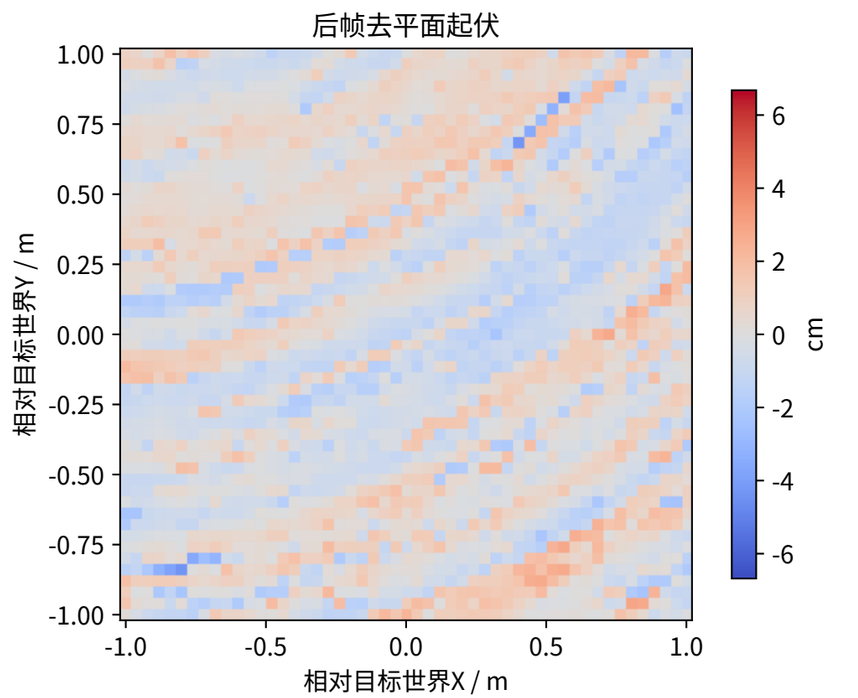
对照：全红→全绿｜样本#818，plus20_moving
世界(-1,12)m，29.71→34.91s，覆盖≥100.0%，按组内中位变化选取

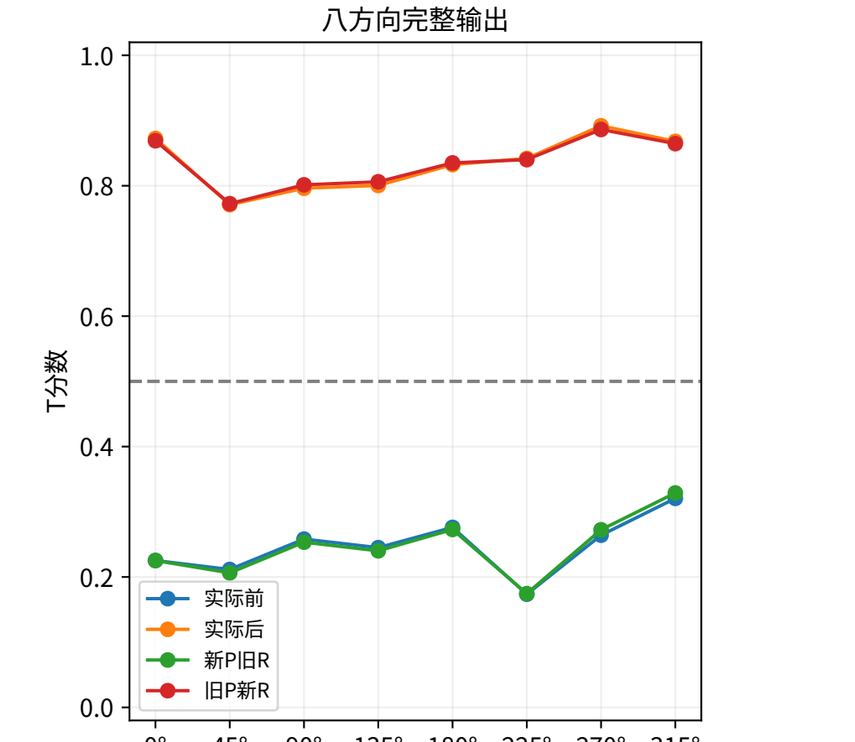




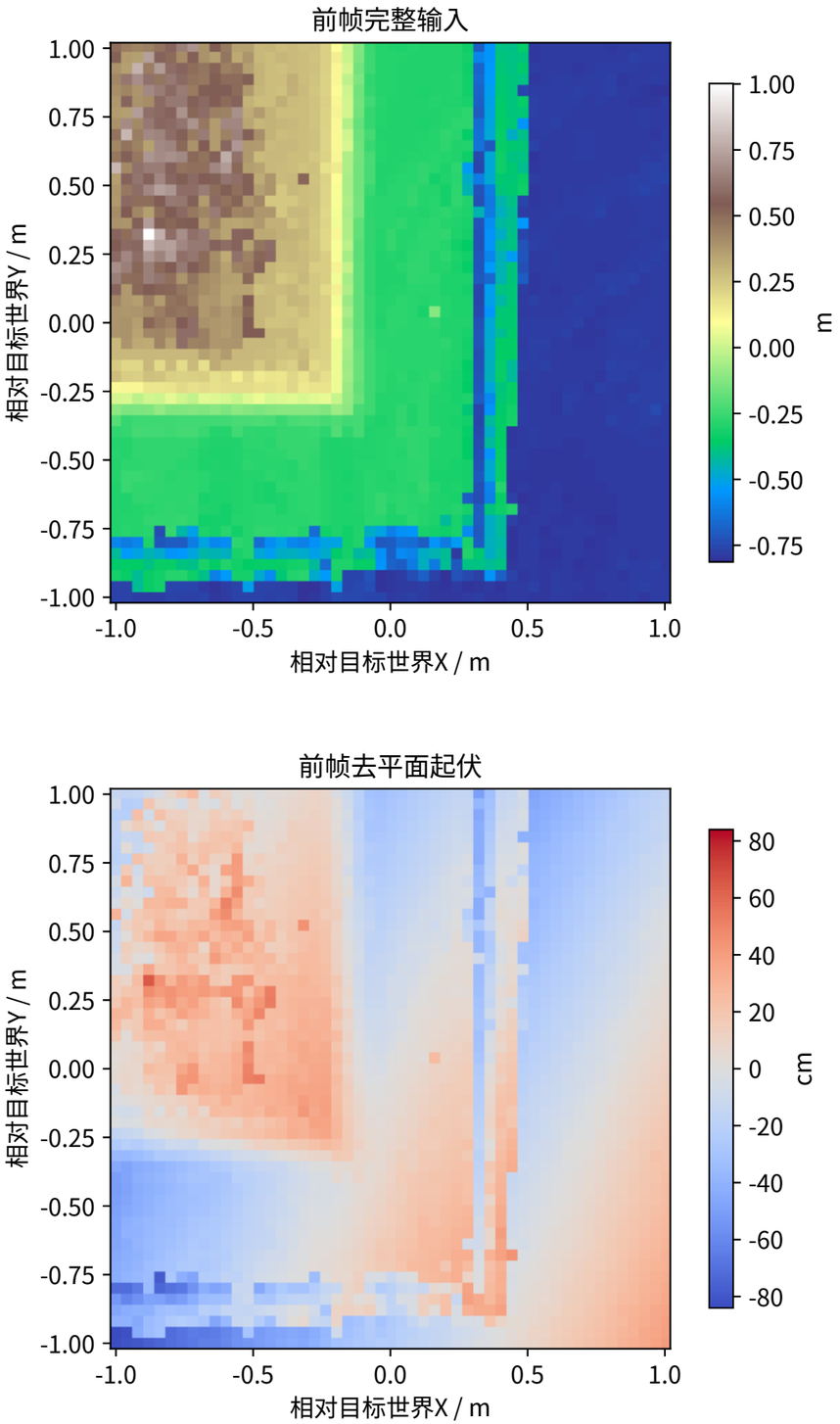


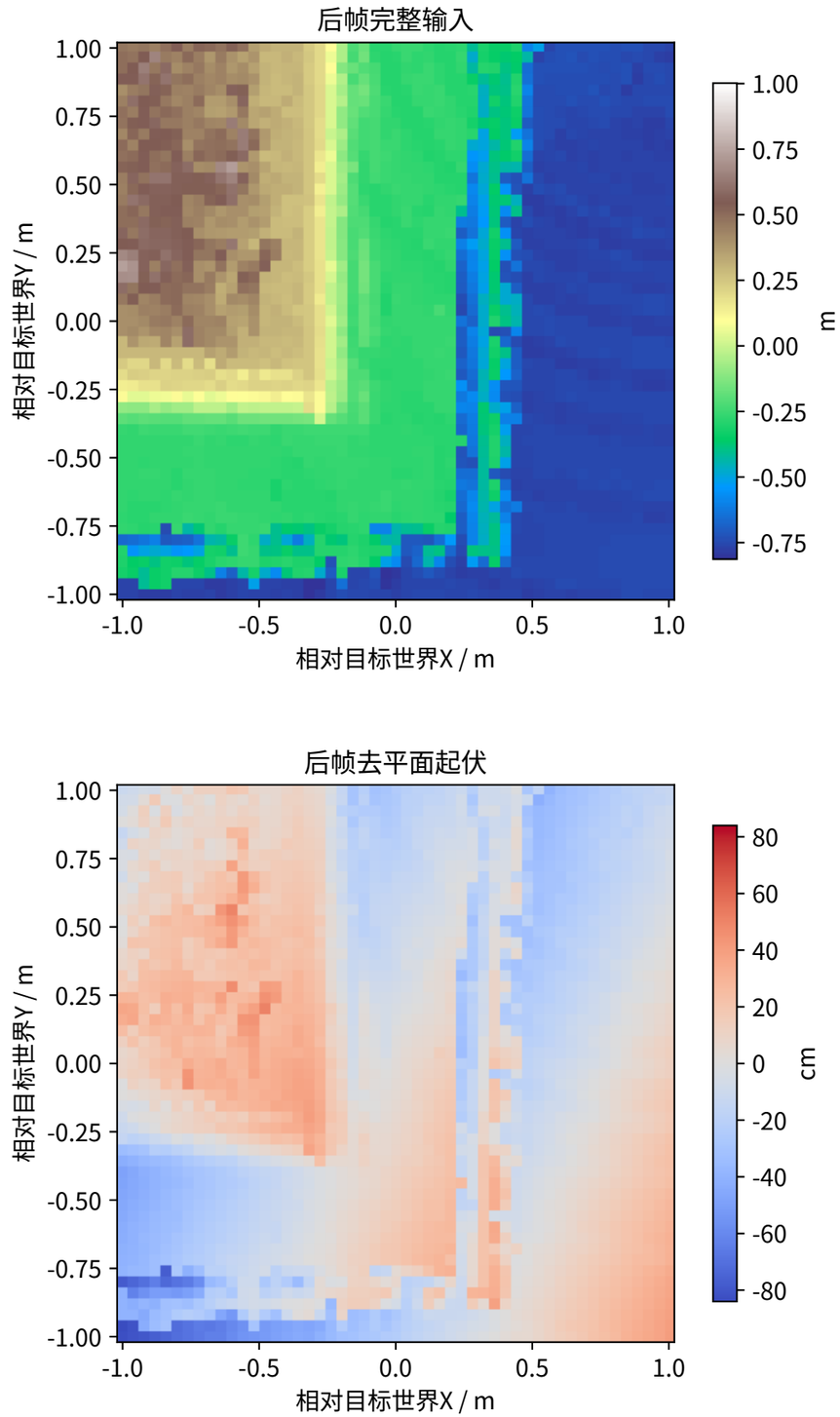


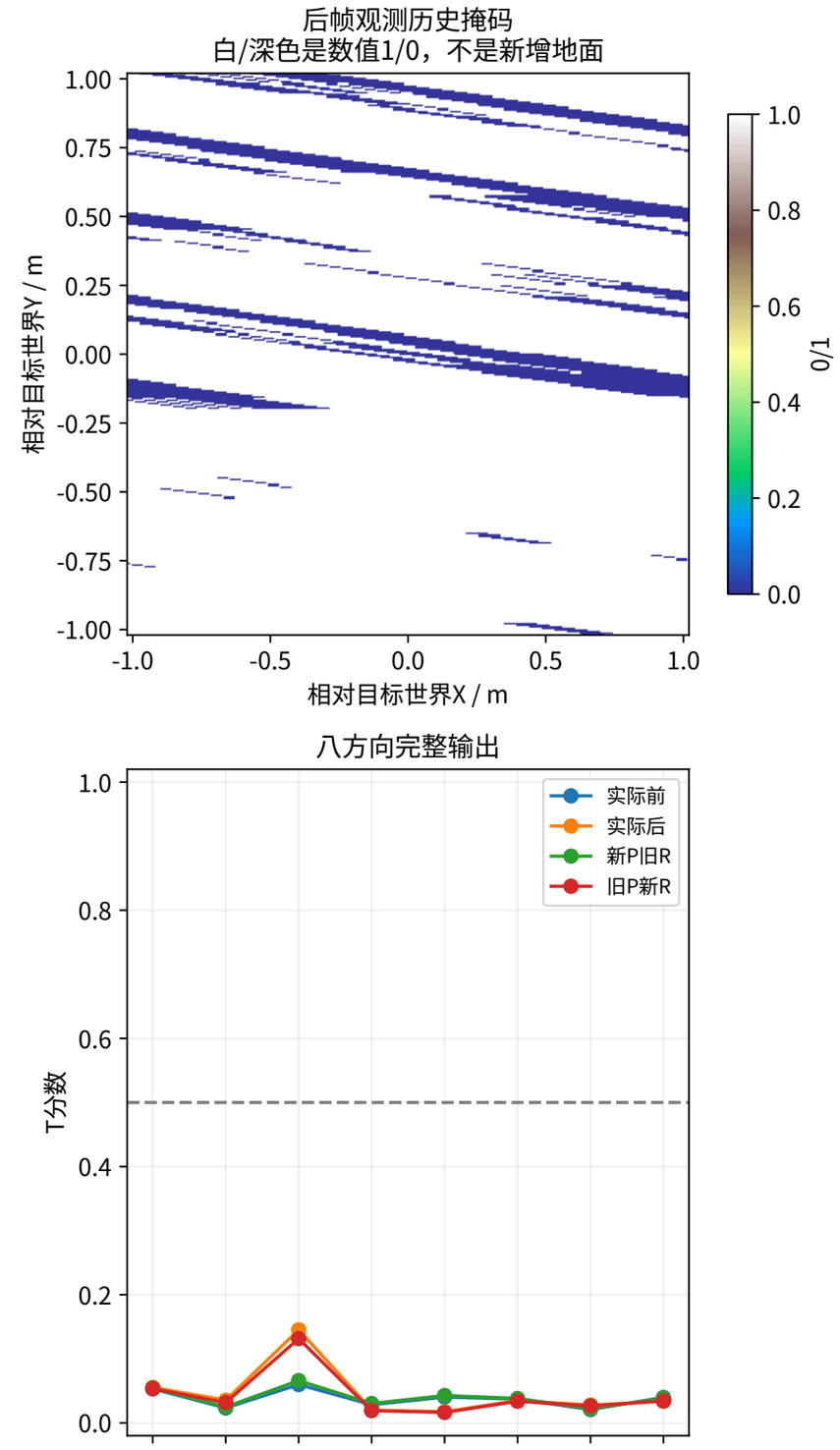




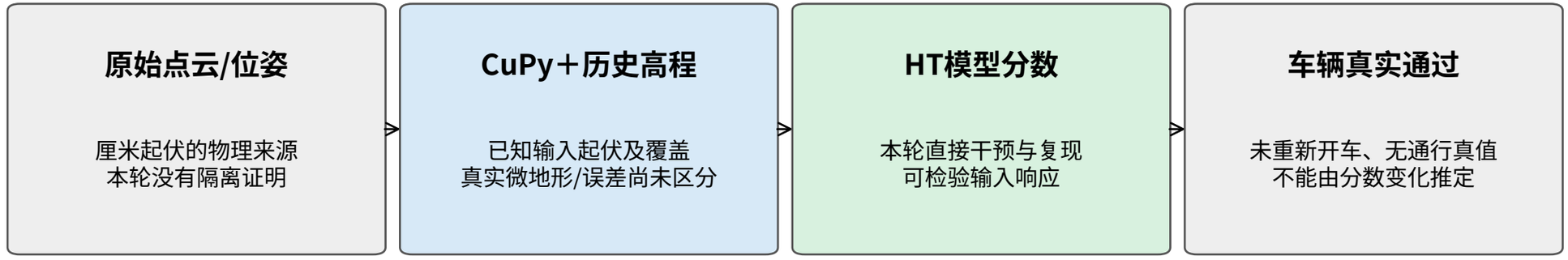
反例：增加起伏，分数反而上升｜样本#706，plus20_moving
世界(-4,15)m，30.30→35.50s，覆盖≥88.7%，明确标记的极端反例，不代表群体比例







示意图③：证据到了哪一层，哪些因果链还没有闭合



已得到的证据：640对主样本中139对平均变化≥0.1，均满足预定的R影响≥2P判据。

需要保留的反证：501/640变化不足0.1；主样本只有1对“八方向全绿→全红”；36/640幅值变化非单调。

公平对照说明：同样1cm高度RMS，平缓倾斜与IID细碎扰动的输出显著不同，空间组织很重要。

不能得出的结论：残差一定是噪声；所有红区都是误报；某阈值已安全；这些窗口等于独立实车实验。

下一步真正要验证：原始点云→高程的误差来源、真实小障碍保持，以及有标签的车辆通行表现。

示意图④：模型首先读取相邻高度差，高度RMS不是唯一决定因素
因此试验支持“空间形态敏感”，不能仅凭1cm扰动就断言模型错误或地面不安全

